Senior Cloud DevOps Engineer – AKS –Bicep –Fully remote – The Aquarium – Raj email urgent approche -19/9/25
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Cloud DevOps Engineer /Software Engineer/Full Stack Developer(Java/.NET/Python) /Sr. Data Engineer/ AI/ML/DL/BI/SharePoint/Dynamics 365 Developer /EDW -Japan (7.5yrs) and Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior Cloud DevOps Engineer /Software Engineer/Sr. Data Engineer /AI/ML Automation Engineer
Responsibilities:

- **Azure & Infrastructure as Code (IaC)**
- Designed, deployed, and managed a resilient and scalable Azure cloud infrastructure using Infrastructure as Code (IaC) methodologies, with a concentrated focus on developing and standardizing reusable Bicep modules such as network.bicep, virtualMachine.bicep, and storageAccount.bicep to ensure consistent, secure, and repeatable deployments.
  E.g.1: module storage 'storageAccount.bicep' = { name: 'storageDeploy' params: { storageAccountName: 'mysa${uniqueString(resourceGroup().id)}' }}
  E.g 2: module vnet 'modules/network/vnet.bicep' = { name: 'vnet-module' params: { location: resourceGroup().location vnetName: 'core-vnet' } }
- Designed and implemented secure Azure landing zones for Linux and Windows workloads, leveraging Azure Blueprints and incorporating security best practices. Integrated these landing zones with M365 services, including MS Teams, SharePoint online, D365, Power Automate, Power App, Power BI) & AI Builder for D365, ensuring secure access and data governance.
- Drove IaC initiatives by implementing Bicep's modularity and utilizing ARM templates to streamline Azure resource deployments, orchestrating resource creation and management through direct API calls to the Azure Resource Manager (ARM) API, which significantly enhanced code consistency and efficiency for critical enterprise infrastructure.
  E.g.: module aksCluster 'modules/compute/aks.bicep' = { name: 'aks-cluster-module' params: { location: resourceGroup().location clusterName: 'enterprise-aks' } }
- Managed resources across local data centers to maintain system availability and scalability, leveraging Azure Arc to unify on-premises infrastructure with the Azure control plane, Azure Monitor for proactive performance and availability monitoring, and Azure Automation for scalable, script-based management.
- Implemented cloud-native architectures with Azure Container Apps/Azure Kubernetes Service (AKS) for containerized microservices and REST APIs, ensuring efficient model integration with enterprise platforms and streamlined data ingestion within an Azure Synapse Analytics-powered lakehouse.
- **Azure IaaS**: Successfully deployed, managed, and maintained Azure IaaS solutions, including compute (Virtual Machines, Scale Sets, AKS), storage (Blob, Queue, Disk), networking (VNet, Load Balancer, VPN Gateway), and security services (Azure AD, Security Center, Key Vault) to support critical business applications.

- **Azure Cloud Optimization**: "Optimized Azure cloud environments for cost efficiency, performance, and security using tools like Azure Cost Management + Billing, Azure Advisor, Azure Monitor, Azure Automation, and Azure Security Center, resulting in 20% cost reduction, 30% performance improvement.
- Managed and configured a range of Azure services, including Virtual Machines, Virtual Networks, Storage Accounts, Azure SQL Database, and Azure Active Directory, to support critical business applications.
- Cloud Infrastructure Engineering: Azure Expertise: Proficient in designing, implementing, and managing Azure infrastructure, including VNets, subnets, NSGs, firewall rules and intrusion prevention systems, SSO, FortiGate Firewalls, FortiClient VPN with MFA, and Azure Entra ID (formerly known as AAD)-based access control for Azure Key Vault.
- **DevOps & CI/CD Automation with AI/ML**
- Architected and managed robust CI/CD processes by integrating a comprehensive suite of DevOps toolsets, including Azure DevOps, Azure Pipelines , GitHub, Jenkins, and YAML, to ensure seamless and automated pipelines.
- Proficient in modern DevOps workflows, utilizing platforms like n8n for workflow automation. Leveraged Git for version control and robust CI/CD practices for continuous integration and deployment, including containerization.
  E.g.1: git push origin main E.g.2: docker build -t myapp:latest . && docker push myapp:latest
- Pioneered the use of Large Language Models (LLMs) as optimizers within CI/CD pipelines to dynamically generate, test, & refactor code, leading to a 25% reduction in deployment time and a significant improvement in code quality across the team.
  E.g., client = AzureOpenAI(azure_endpoint = os.getenv("AZURE_OPENAI_ENDPOINT"),
  api_key=os.getenv("AZURE_OPENAI_API_KEY"), api_version="2024-02-15-preview")
  response = client.chat.completions.create(model="gpt-4o",
  messages=[{"role": "user", "content": "Refactor this code to optimize performance: {code_snippet}"}])
  llm_output = response.choices[0].message.content
- **Operations, Infrastructure, & API Management**
- Supported system and operations integrations by engineering robust API and file transfer setups using both native cloud services & scripting. Implemented secure API integrations via PowerShell with Invoke-RestMethod & Invoke-WebRequest & managed file transfers using command-line tools like azcopy for high-throughput data migration& scp for secure file copying.
- Oversaw infrastructure and network management tasks, including automated system patching, middleware upgrades, and proactive monitoring using Azure tools like Azure Automation, Azure Update Management, and Azure Monitor. Developed and executed PowerShell scripts to ensure consistent and compliant configurations across the environment.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior Cloud DevOps Engineer/Senior Data Engineer /Python /ML Developer
Responsibilities:

- **GCP Cloud Infrastructure & DevOps**
- Engineered and deployed resilient, scalable GCP cloud infrastructure for financial trading systems, leveraging IaC with a strong focus on GCP Deployment Manager templates. Standardized reusable templates for core services like VPCs, Compute Engine instances, and Cloud Storage buckets to ensure consistent, secure, and repeatable deployments.
  E.g., gcloud deployment-manager deployments create trading-network --config network-vpc.yaml
- Designed and implemented secure GCP projects for both Linux and Windows workloads using GCP IAM policies, integrating them with internal enterprise services to ensure secure access and data governance.
  E.g., gcloud deployment-manager deployments create security-policies --config iam-policies.yaml
- Architected foundational GCP infrastructure including VPCs, subnets, and Compute Engine instances for Docker containers using Deployment Manager for its declarative syntax and seamless integration with GCP services.
  E.g., gcloud deployment-manager deployments create trading-cluster --config docker-cluster.yaml
- Spearheaded IaC initiatives by building a modular and scalable Deployment Manager framework, adhering to CI/CD principles to automate and streamline deployments.
  E.g., gcloud deployment-manager deployments create secrets-manager --config secrets-manager.yaml
- Managed hybrid cloud resources across local data centers using GCP Cloud VPN and Dedicated Interconnect to unify on-premises infrastructure with GCP. Utilized GCP Stackdriver for proactive performance monitoring and Cloud Functions for scalable, event-driven automation. E.g., gcloud compute vpn-gateways create onprem-vpn
- Implemented cloud-native architectures with GCP Compute Engine and Docker for containerized microservices and REST APIs, ensuring seamless integration with enterprise platforms and efficient data ingestion into a BigQuery-powered data warehouse. E.g., gcloud compute instances create trading-api-instance
- **CI/CD Automation & AI/ML-Powered Automation & Insights**
- Developed self-learning Multi-agent systems (MAS), a subfield of distributed AI in Python to automate and optimize algorithmic strategies, enabling models to autonomously adapt to dynamic financial market patterns.
- Engineered intelligent rule-based Robotic Process Automation (RPA) workflows powered by NLP libraries (e.g., NLTK, spaCy) for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Architected and managed robust CI/CD processes for RBA financial trading systems by integrating a comprehensive suite of DevOps toolsets, including Azure DevOps, Jenkins, and YAML. This ensured seamless and automated pipelines from code commit to production deployment. Collaborated within an Agile framework to enhance DevOps efficiency.
- Developed and implemented fully automated CI/CD pipelines using Azure DevOps Pipelines and GitHub Actions to automate builds, tests, and deployments.

- Orchestrated automated workflows using Robotic Process Automation (RPA) tools to streamline repetitive tasks across the development lifecycle, ensuring consistency and accuracy in a pre-AI environment. Utilized Git for version control and robust CI/CD practices for continuous integration and deployment. E.g., rpa-cli run --task "deploy-to-dev-env"
- Pioneered the use of rule-based RPA scripts to automate critical steps within the CI/CD pipeline, such as code validation and deployment triggers. This led to a significant reduction in manual errors and accelerated deployment cycles.
  E.g., powershell -file ".¥deploy_to_uat.ps1" --environment uat
- **Operations, Infrastructure, & API Management**
- Managed and secured API integrations and data transfers for a financial trading platform. Engineered robust API and file transfer setups, using Apigee to govern access and enforce policies. Implemented secure data movement via gcloud CLI with gcloud storage cp for high-throughput migration and scp for secure file copying.
  E.g., gcloud apigee apis create --api=trading-api --org=my-org gcloud storage cp gs://source-bucket/data.csv .
- Oversaw comprehensive infrastructure and network management, including automated system patching and middleware upgrades. Leveraged GCP Cloud Functions and Stackdriver Logging to ensure consistent and compliant configurations across the environment. E.g., gcloud functions deploy auto-patch-vm --trigger-topic=patching-topic
- Established proactive monitoring and alerting using GCP Cloud Monitoring, ensuring high availability and performance of all trading systems. Developed and executed gcloud CLI and shell scripts to automate routine operational tasks and maintain configuration compliance. E.g., gcloud monitoring dashboards create --config-from-file=dashboard.json
  gcloud compute ssh my-instance --command "sudo apt-get update"
- Drove efficiency by implementing Deployment Manager and gcloud CLI to orchestrate resource creation through direct calls to the GCP APIs, significantly enhancing code consistency for critical enterprise infrastructure.
  E.g., gcloud deployment-manager deployments create automated-reporting --config reporting-cloud-function.yaml
- Application Development & Backend APIs: Developed and deployed scalable applications using MuleSoft Anypoint Platform, including Mule 4 and Anypoint Studio 7.x, with a strong focus on robust backend API development and seamless system integrations. **API Development**: POST /api/v1/orders HTTP/1.1 { "item": "widget", "qty": 2 }
  System Integration: Flow orchestrates Salesforce, SAP, and custom REST services for order processing.
- Engineered scalable, high-performance Python applications and robust backend APIs, facilitating seamless cloud-native integrations across GCP, Azure, and AWS services for real-time data processing and AI solution enablement.
  E.g., sqs_client.send_message(QueueUrl=queue_url, MessageBody=json_data)
  pubsub_client.publish(topic_path, json_data.encode("utf-8"))

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Sr. Cloud DevOps Engineer/Sr. Data Engineer/Python/ML/Java/Mulesoft Developer
Responsibilities:

- **AWS Cloud Infrastructure & DevOps**
- Engineered and deployed resilient, scalable AWS cloud infrastructure for financial trading systems, leveraging Infrastructure as Code (IaC) with a strong focus on AWS CloudFormation templates. Standardized reusable templates for core services like VPCs, EC2 instances, and S3 buckets to ensure consistent, secure, and repeatable deployments.
- aws cloudformation create-stack --stack-name trading-network --template-body file://network-vpc.json
- Designed and implemented secure AWS landing zones for both Linux and Windows workloads using AWS Service Catalog and custom IAM policies, integrating them with internal enterprise services to ensure secure access and data governance.
- aws cloudformation create-stack --stack-name security-policies --template-body file://iam-policies.json
- Architected foundational AWS infrastructure including VPCs, subnets, and EC2 Container Service (ECS) clusters using CloudFormation for its declarative syntax and seamless integration with AWS services.
  E.g., aws cloudformation create-stack --stack-name trading-ecs-cluster --template-body file://ecs-cluster.json
- Spearheaded IaC initiatives by building a modular and scalable CloudFormation framework, adhering to CI/CD principles to automate and streamline deployments.
- E.g., aws cloudformation create-stack --stack-name secrets-manager --template-body file://secrets-manager.json
- Drove efficiency by implementing CloudFormation and AWS CLI to orchestrate resource creation through direct calls to the AWS APIs, significantly enhancing code consistency for critical enterprise infrastructure.
- E.g., aws cloudformation create-stack --stack-name automated-reporting --template-body file://reporting-lambda.json
- Managed hybrid cloud resources across local data centers using AWS Storage Gateway and Direct Connect to unify on-premises infrastructure with AWS. Utilized Amazon CloudWatch for proactive performance monitoring and AWS Lambda for scalable, event-driven automation. E.g., aws storagegateway create-gateway --gateway-name onprem-gateway
- Implemented cloud-native architectures with Amazon EC2 Container Service (ECS) for containerized microservices and REST APIs, ensuring seamless integration with enterprise platforms and efficient data ingestion into an Amazon Redshift-powered data warehouse. E.g., aws ecs create-service --service-name trading-api --cluster trading-ecs-cluster
- Architected and implemented hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Configured encrypted tunnels for secure data transfer using IPsec via Cloud VPN Gateways across all three platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with AWS CloudWatch.

- Integrated M365 environments (O365, MS Teams, SharePoint online, Dynamics 365, Power Automate,Power App, Power BI) & AI Builder for D365 into the AWS Landing Zones for a comprehensive and secure platform for Microsoft workloads.
- **CI/CD Automation & ML-Powered Automation & Insights**
- Engineered and deployed high-performance Machine Learning models within Python applications, leveraging Azure Machine Learning Studio's capabilities with R and Python support for advanced predictive modeling and synthetic data generation, significantly impacting financial applications. E.g., model_output <- predict(trained_model, newdata)
- Developed robust, data-driven pipelines for automating AWS IaC pipeline creation, prediction, and optimization using SageMaker, CloudFormation, and boto3 (e.g., cfn = boto3.client('cloudformation') …).
- **Operations, Infrastructure, & API Management**
- Managed and secured API integrations and data transfers for a financial trading platform. Engineered robust API and file transfer setups, leveraging Azure's native cloud services and scripting. Implemented secure data movement via PowerShell with Invoke-RestMethod and azcopy for high-throughput migration and scp for secure file copying.
- Oversaw comprehensive infrastructure and network management, including automated system patching and middleware upgrades. Leveraged Azure Automation and Azure Update Management to ensure consistent and compliant configurations across the environment.
- Established proactive monitoring and alerting using Azure Monitor, ensuring high availability and performance of all trading systems. Developed and executed PowerShell scripts to automate routine operational tasks and maintain configuration compliance.
- Managed and optimized cloud-based applications for the student enrollment system, ensuring high availability, scalability, and cost-effectiveness on leading cloud platforms.
  E.g., aws s3 sync local_data/ s3://uc-student-data-bucket/ --delete (Ensures data consistency and efficient storage)

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Python Application Developer /DevOps Engineer/Data Engineer /Software Test Engineer
Responsibilities:

- **Data-driven Python Application & DevOps**
- Maintained a DevOps-centric approach to Python application development, focusing on building high-performance solutions for real-time data processing and analytics.

- **AWS Cloud & Infrastructure**
- Architected and implemented a modern 3-layer lambda architecture on AWS, utilizing a range of AWS services for real-time and batch data processing.
  E.g., aws cloudformation create-stack --stack-name automated-reporting --template-body file://reporting-lambda.json
- **CI/CD Automation & ML-Powered Automation**
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation and optimization, leveraging SageMaker, CloudFormation, and boto3. E.g., template = update_template_based_on_predictions(template, predictions)
- Applied these capabilities to develop machine learning models that accurately assess the likelihood of specific claim outcomes or potential fraud indicators, enabling proactive risk management and optimized resource allocation in the injury compensation domain. E.g.: high_risk_claims = claims[model.predict_proba(features)[:, 1] > 0.7]
- Machine Learning Models: Developed and deployed advanced predictive models using Python and Azure Machine Learning, optimizing claims processing efficiency and enhancing fraud detection capabilities. E.g, model_output = trained_model.predict(new_claim_features)
- **Architected and implemented a modern 3-layer lambda architecture on AWS**:
- **Batch layer**: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed AWS Redshift with its built-in data processing capabilities for near-real-time processing & analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer/Data Engineer /Software Test Engineer/SharePoint Consultant
Responsibilities:

- Architected and built scalable cloud-native web applications using AWS services and Python.
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda. Work with Amazon DynamoDB for data storage and retrieval
- Implement APIs and integrate with various systems using API Gateway
- Deploy and maintain web applications on Amazon Web Services (AWS) cloud platform
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **Security and Compliance**: **Implement security best practices**: Regularly patch systems and applications.
- **GCP Stack Proficiency:** Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages).
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of continuous integration (CI) and automated deployment processes, leading to a significant reduction in software feature delivery time.
- Utilized scripting languages and configuration management tools to implement Infrastructure as Code (IaC) principles, enhancing deployment processes and scalability.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Administered and monitored Unix/Linux-based systems, ensuring the smooth operation of critical applications.
- Provided Level 2/3 technical support for all Microsoft infrastructures, including Active Directory (AD), DNS, DHCP, Exchange 2010, and SharePoint.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running costs.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Served as Project Manager to replace the existing insurance web application (from IBM AS/400 and LAMP) with a WISA-based system, significantly reducing IT operational costs and integrating with a Salesforce Automation (SFA) platform for improved marketing results.
- Planned, implemented, and supported all core IT systems including Hyper-V, VMware ESX & vSphere 4.x, HP/EMC/Dell SAN Storage, and Windows Server 2008, and applied security principles and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for desktop productivity suite issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Designed data warehouse architecture, including hardware infrastructure, schema definitions, and Fact and Relational Dimensions tables.
- Developed data warehouse and business reporting solutions to meet stakeholder requirements using SSIS, SSAS, and SSRS.
- Used SSIS ETL tools to extract and transform complex data, loading it into the data warehouse to produce Analysis Services Cubes for business reporting via SSRS or Data Marts to serve business applications.
- Created and deployed reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio, including debugging, monitoring, and troubleshooting BI solutions.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Managed and maintained company databases, ensuring data accuracy and integrity.
- Defined the target warehouse module, and created target user, module, and schema.
- Built and deployed machine learning models using k-nearest neighbors (k-NN) for customer churn prediction and fraud detection, achieving a 15% improvement in accuracy.
- Involved in UAT and TEST in a QA development environment by writing VBScript and batch files to automate processes.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI (ChatGPT-4o, Multi-GPTs, Auto-Agents), Amazon Bedrock (Claude/Titan) & SageMaker, Azure OpenAI Service, Azure ML, Azure AI Studio, Copilot, Azure AI Foundry, Google Vertex AI, Gemini API (formerly MakerSuite) , Agent-2-Agent
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
- **Workflow & Automation Platforms**: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve

Senior AI Engineer– Bedrock-SageMaker Specialist CBA –Rackspace - 9/9/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior AI Engineer– Bedrock-SageMaker Specialist / GenAI | AI Build | Automation Engineer/ Snr DevOps /Data Engineer
Principle Software Engineer/Full Stack Developer(Py/Java/.Net/Mulesoft /AI/ML/DL/BI/SP/D365)-Japan (7.5yrs) & Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), **Sydney**
Gov. Health: Senior AI Engineer–Bedrock-SageMaker Specialist/ GenAI | Principle Software Engineer | Automation Engineer
Responsibilities:

- **Agentic Platform Development & Generative AI**
- Architected and deployed end-to-end AI systems from experimentation to production, applying practical ML techniques such as RAG, fine-tuning, NLP, vectorisation, and KNN to solve real-world business use cases.
  E.g., model = PeftModel.from_pretrained(model, peft_model_id).
- Engineered and optimized end-to-end Retrieval-Augmented Generation (RAG) pipelines, including advanced Graph RAG and intelligent chunking with LangChain, to enhance contextual understanding and answer complex semantic queries, boosting performance and reducing API calls by over 20%.
  E.g., text_splitter = RecursiveCharacterTextSplitter(chunk_size=1000, chunk_overlap=200).
  E.g. 2: async def query_llm_with_rag(index: VectorStoreIndex, question: str) -> Dict[str, Any]:
- Developed and implemented an advanced Graph-R1 agentic RAG framework, integrating structured knowledge graphs and reinforcement learning to enhance multi-hop reasoning and achieve a 25% improvement in the accuracy of semantic query responses for a knowledge-intensive platform.

- Developed and orchestrated sophisticated autonomous AI agents using frameworks like LangGraph, AutoGen, and Semantic Kernel, and leveraging inter-agent communication protocols and tools like n8n to integrate with external APIs, driving operational efficiency and intelligent decision-making.
  E.g, response = await kernel.invoke_async("email_plugin.summarize", text=email_body).
- Architected and deployed enterprise-grade Generative AI solutions using Amazon Bedrock, leveraging a suite of pre-trained foundation models including Anthropic's Claude 3 for its long context window and Amazon Titan. Applied prompt engineering and LLM-based RAG to build a context-aware customer support chatbot, reducing response time by 30%.
- Utilized cutting-edge Graph-R1 and Reinforcement Learning to build a robust Agentic Graph RAG framework, successfully reducing hallucination rates by configuring security-level parameters.
- Developed a comprehensive observability framework for data platforms, leveraging insights from CloudWatch Logs to predict ETL failures in Airflow and Glue jobs and implementing automated remediation patterns to reduce incident response time by over 40%. E.g., boto3.client('lambda').invoke(FunctionName='AutomatedRemediationFunction', InvocationType='Event', Payload='{"failure_type": "ETL_Error", "job_id": "job_123"}')
- Consistently delivered high-quality AI solutions within agile sprint cycles, using data-driven metrics like MTTR and FPR to evaluate performance and iterate on improvements, which allowed for effective communication of complex AI logic to non-technical stakeholders via dashboards and chat interfaces.
- **AI-Powered Automation & Insights**
- Engineered intelligent RPA workflows powered by NLP for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Developed scalable Python applications, leveraging Generative AI frameworks (LangChain for LLM orchestration, LlamaIndex for RAG) and speech-to-text models like Whisper to automate the transcription, summarization, and analysis of customer calls. Optimized deployment of these systems on AWS (using Bedrock and OpenAI frameworks) while integrating with core business systems via data-intensive APIs (e.g., Transactions API, Balances API) and enterprise data platforms.
- Integrated MongoDB Atlas as a vector database for RAG (Retrieval-Augmented Generation), enabling real-time and context-aware enterprise insights.
- Developed an AI-powered insights platform for a major financial institution by creating a knowledge graph to map complex customer and transaction data. Used embeddings to represent this information as high-dimensional vectors, which were then stored in a vector database (Pinecone/Weaviate) to enable lightning-fast semantic search and relationship discovery, enhancing fraud detection and personalized financial recommendations.
- Developed an intelligent AI agent using an Amazon Scraper (Python, requests, BeautifulSoup, pandas) with Streamlit UI, demonstrating robust data acquisition and constraints bypassing.
- **Machine Learning & Model Development**
- Developed, trained, and deployed a suite of machine learning and deep learning models on Amazon SageMaker using PyTorch, TensorFlow, and Hugging Face Transformers, optimizing data pipelines and automating model retraining to improve predictive accuracy and deliver NLP solutions.
- Built robust predictive models and secure AI APIs using diverse machine learning algorithms (KNN, CNN, RNN, ANN, GAN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning), applying Scikit-learn and Keras to drive Next Best Action (NBA) recommendations and pattern discovery.
- Drove the full lifecycle of AI solutions, from ideation and opportunity identification with key stakeholders to the design and delivery of scalable prototypes. Conducted comprehensive experiments to evaluate model performance, iterating on design to optimize outcomes and ensure compliance with ethical AI standards.

- **DevOps, Cloud Infrastructure & Backend Development**
- Pioneered the use of LLMs and agentic workflows to automate and refactor code within scalable CI/CD pipelines, reducing deployment time by 25%. Engineered and deployed mission-critical Python applications and GenAI platforms on a multi-cloud architecture (AWS, Azure, GCP), leveraging IaC, MCP+RAG, and n8n orchestration.
  - E.g., llm_output = gemini.generate_text(f"Refactor this code to optimize performance: {code_snippet}")
  - E.g., ansible-playbook provision_rhel_app_servers.yml
  - E.g., requests.post(f"{api_url}/api/v1/deployments", json={'app':'genai-service', 'version':'v1.0.0'}).
- Leveraged AWS Kiro's spec-driven development & agentic capabilities to design, build, and maintain scalable, secure AWS infrastructure (EC2, EKS, S3, RDS, Lambda, SageMaker, Amazon Neptune) for hosting advanced Graph RAG pipelines, vector databases, & LLMs, Using AWS Bedrock for diverse foundational models (e.g., Amazon, Anthropic, AI21 Labs).
- Implemented cloud-native architectures with AWS ECS/EKS for containerized microservices and REST APIs, ensuring efficient model integration with enterprise platforms streamlined data ingestion within an AWS Redshift-powered lakehouse.
- Designed and developed scalable backend APIs and seamless integrations using Python frameworks (Django, Flask), optimized data pipelines for AI model inference, enabling real-time predictions on large datasets.
- Pioneered the use of Large Language Models (LLMs) as optimizers within CI/CD pipelines to dynamically generate, test, and refactor code, leading to a 25% reduction in deployment time and a significant improvement in code quality across the team. E.g., llm_output = bedrock_client.invoke_model(modelId='anthropic.claude-v3', or 'amazon.titan-text-express-v1'
  body={ "prompt": "Refactor this code to optimize performance: {code_snippet}"
- **Technical Leadership & Innovation**

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- Led end-to-end GenAI solution development from secure audio ingestion, automated transcription, LLM orchestration, RAG pipeline integration, to intelligent summarization—ensuring seamless data flow across multi-cloud.
- Leveraged a self-recursive Artificial Superintelligence Architecture (ASI-ARCH) on the Harrison Platform (Nvidia DGX-1 systems) to autonomously innovate and optimize AI models for financial management. Enhanced algorithmic trading strategies and risk mitigation by discovering emergent market principles beyond human intuition.
- Integrated M365 environments (O365, MS Teams, SharePoint online, Dynamics 365, Power Automate,Power App, Power BI) & AI Builder for D365 into the AWS Landing Zones for a comprehensive and secure platform for Microsoft workloads.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov Central Bank: Senior AI/ML Engineer/Automation Engineer/Snr DevOps/Data Engineer /AI/ML/DL/Mulesoft Developer
Responsibilities:

- AI/ML-Powered Automation & Insights
- Developed self-learning Multi-agent systems (MAS), a subfield of distributed AI in Python to automate and optimize algorithmic strategies, enabling models to autonomously adapt to dynamic financial market patterns.
- Engineered intelligent rule-based Robotic Process Automation (RPA) workflows powered by NLP libraries (e.g., NLTK, spaCy) for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Developed scalable Python applications, leveraging frameworks like scikit-learn and NLTK for data indexing and natural language processing.E.g., model.predict(vectorizer.transform(["RBA interest rate hike triggers AUD volatility"]))
- Led end-to-end machine learning solution development from secure audio ingestion, automated transcription, to intelligent summarization—ensuring seamless data flow across multi-cloud.Integrated speech-to-text APIs for automatic high-volume call transcription and sentiment analysis, directly logging customer sentiment into CRM workflows.
- Developed computer vision models using OpenCV & Scikit-image to integrate image recognition for automated validation of property documents & anomaly detection, streamlining regulatory workflows & enhancing decision-making accuracy.
- Technical Foundations & Modeling
- Built robust predictive models and secure machine learning APIs using diverse algorithms (KNN, CNN, RNN, ANN, GAN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning), applying Scikit-learn and Keras to drive next best action (NBA) recommendations and pattern discovery.
- Leveraged PyTorch and TensorFlow to develop and deploy deep learning models, including fine-tuning pre-trained models for NLP and analytics.Utilized Nvidia DGX-1 systems to develop and optimize machine learning models for financial management.
- Enhanced algorithmic trading strategies and risk mitigation by discovering new market principles beyond human intuition.Engineered scalable, mission-critical production applications in Python, driving robust system integrations and continuous deployment for machine learning services with core code such as requests.post(f"{api_url}/api/v1/deployments", json={'app':'ml-service', 'version':'v1.0.0'}).
- DevOps & Cloud Infrastructure
- Automated the creation of secure, scalable IaC CI/CD pipelines and isolated machine learning platforms, enabling robust deployment of Python (Django, Flask) and other server-side applications across AWS, Azure, and GCP.
- Implemented cloud-native architectures with AWS ECS for containerized microservices and REST APIs, ensuring efficient model integration with enterprise platforms and streamlined data ingestion within a data lakehouse.
- Specialized in designing and implementing end-to-end automation solutions using Python & Red Hat Ansible Automation Platform (including Ansible Tower and AWX), developing modules and plugins to streamline secure, low-latency infrastructure provisioning for RBA trading platforms across hybrid cloud, on-prem, and RHEL9 environments. E.g., ansible-playbook deploy_low_latency_trading_engine.yml to automate trading engine deployments.
- Leveraged PagerDuty as a central incident management system, effectively aggregating and transforming alerts from diverse monitoring and observability platforms (e.g., Airflow and Glue ETL failures, Datadog, Prometheus, AWS services) into actionable incidents.
- Integrated PagerDuty with Amazon DevOps Guru to leverage its AIOps/ML-powered operational insights, translating them into actionable incidents, triggering a PagerDuty incident. Runbook Automation via Bamboo: Reduce support costs by 50% by eliminating toil with PagerDuty. E.g., aws cloudwatch put-metric-data --metric-name CriticalApplicationError --namespace MyAIOpsApplication --value 1 --unit Count --dimensions Environment=Production,Service=AuthService.
- Developed and deployed scalable applications using MuleSoft Anypoint Platform, including Mule 4 & Anypoint Studio 7.x, with a strong focus on robust backend API development and seamless system integrations.E.g. 1: POST /api/v1/orders HTTP/1.1 { "item": "widget", "qty": 2 }E.g. 2: Flow orchestrates Salesforce, SAP, and custom REST services for order processing.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr ML Engineer/Automation Engineer /Integration Solution Architect /ML/DL/Mulesoft Developer
Responsibilities:

- **Automated Systems and Machine Learning**

- Engineered self-learning Multi-agent systems (MAS) in Python to automate the management of student enrollment and financial data, enabling models to autonomously adapt to complex patterns for enhanced system performance.
- Engineered intelligent rule-based Robotic Process Automation (RPA) workflows powered by NLP libraries (e.g., NLTK, spaCy) for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Developed computer vision models (using OpenCV and Scikit-image) to integrate image recognition for the automated analysis of visual data and anomaly detection, streamlining data processing and enhancing decision-making accuracy.
- Led end-to-end machine learning project development, from data collection and automated transcription to statistical analysis and intelligent summarization, ensuring efficient data flow for research and academic projects.
- Developed scalable Python applications, leveraging core frameworks like scikit-learn, NLTK, Theano, and Caffe for data preprocessing, modeling, and natural language processing. Utilized third-party speech-to-text services for automatic transcription and sentiment analysis, integrating results into custom-built data analysis pipelines.
- **Machine Learning and Data Insights**
- Built robust predictive models and secure APIs using diverse machine learning algorithms (KNN, CNN, RNN, ANN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning). Applied Scikit-learn and Keras for projects focused on pattern discovery and predictive analytics.
- Utilized GPU-accelerated systems to develop and optimize machine learning models for research projects. Developed and tested algorithmic strategies by applying emerging principles in statistical analysis.Integrated databases like PostgreSQL or MongoDB for managing and analyzing data in projects, enabling real-time and context-aware insights.
- **Systems Engineering and Automation**
- Implemented cloud-based architectures with AWS ECS for containerized microservices and REST APIs, ensuring efficient model integration and streamlined data ingestion.
- Automated the creation of secure and scalable infrastructure for machine learning experimentation using Infrastructure-as-Code (Terraform) and automated CI/CD pipelines (e.g., Jenkins) for robust deployment across cloud platforms like AWS, Azure, and GCP.
- Used AWS infrastructure (EC2, S3, and RDS) to design, build, and maintain scalable systems for hosting ML models.
- Designed and developed scalable backend APIs and integrations using Python frameworks (Django, Flask), optimizing data pipelines for efficient machine learning model training and enabling near real-time predictions on datasets.
- Developed an intelligent web scraper (Python, requests, BeautifulSoup, pandas) with a simple UI built using Django or Flask to demonstrate robust data acquisition and processing capabilities.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: ML Engineer /Automation Engineer/Software Test Engineer /DevOps/Data Engineer
Responsibilities:

- **Application Development and Machine Learning**
- Developed self-learning multi-agent systems in Python to automate and optimize the processing of injury claims, enabling autonomous adaptation to claims data and improving overall system efficiency.
- Developed and deployed predictive models using Python libraries like scikit-learn to accurately assess the likelihood of specific claim outcomes or potential fraud indicators, enabling proactive risk management and optimized resource allocation in the injury compensation domain.
- Python Application Developer with a robust quantitative foundation, adept at building high-performance solutions. Proficient in Python and Java, with capabilities to design, develop, and optimize ML applications and deliver actionable insights.
- Engineered secure and scalable backend services using Python (e.g., Flask) and Java to efficiently manage high volumes of injury compensation claim data, ensuring data integrity and seamless integration with critical EML systems.
- Designed and implemented user interfaces with jQuery and JavaScript, optimizing the claimant experience for EML's injury compensation portal and streamlining data presentation for complex claim workflows.
- Extensive expertise in programming languages such as Java, Python, C#, and PowerShell scripting languages.Data Engineering & ArchitectureArchitected and implemented data solutions on AWS, focusing on traditional data warehousing and ETL processes.
- Utilized Amazon EMR for batch processing of large data volumes, and AWS Redshift for near-real-time data analysis.
- Employed Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Designed and developed data solutions using AWS services like EC2, RDS, and S3.Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created and implemented mapping documents that defined how source data fields corresponded to target data fields, including writing code to map the data according to defined rules.
- DevOps & Automation

- Automated deployment of scalable machine learning applications and services, leveraging AWS services like EC2 to ensure high availability.Championed continuous integration and delivery (CI/CD) pipelines using tools like Jenkins for secure, continuous application delivery.
- Utilized PowerShell scripting to automate system configurations, testing, and deployment processes.Followed software development best practices for versioning, deployment, testing, and automation.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps/Data Engineer /Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Software Development and Deployment: Architected and built scalable web applications using Python and deployed them on traditional servers.
- Infrastructure Management: Managed infrastructure using scripting languages and automation tools.
- Application Development: Developed applications using Python for data processing and business logic.
- Data Management: Utilized relational databases for data storage and retrieval.
- System Integration: Implemented APIs and integrated with diverse systems.
- Automated build and deployment processes using Jenkins and shell scripting, managing server configurations with Puppet to ensure reliable delivery of machine learning models for research projects.
- Deployed and maintained web applications on virtual servers. Managed infrastructure using shell scripts and other automation tools, integrating them into CI/CD pipelines (e.g., Jenkins). Developed REST APIs using Python and Java.
- Monitored server resources using on-premise tools and implemented logging, monitoring, and security measures.
- **Security and Compliance**: Implemented security best practices and regularly patched systems and applications.
- **Agile and Data Expertise**: Experienced working in agile environments with strong data engineering practices.
- Other Proficiencies: Possessed a strong understanding of SQL, Python scripting, and SharePoint.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of continuous integration (CI) and automated deployment processes, leading to a significant reduction in software feature delivery time.
- Utilized scripting languages and configuration management tools to implement Infrastructure as Code (IaC) principles, enhancing deployment processes and scalability.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Administered and monitored Unix/Linux-based systems, ensuring the smooth operation of critical applications.
- Provided Level 2/3 technical support for all Microsoft infrastructures, including Active Directory (AD), DNS, DHCP, Exchange 2010, and SharePoint.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running costs.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Served as Project Manager to replace the existing insurance web application (from IBM AS/400 and LAMP) with a WISA-based system, significantly reducing IT operational costs and integrating with a Salesforce Automation (SFA) platform for improved marketing results.
- Planned, implemented, and supported all core IT systems including Hyper-V, VMware ESX & vSphere 4.x, HP/EMC/Dell SAN Storage, and Windows Server 2008, and applied security principles and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for desktop productivity suite issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for Business Intelligence (BI) solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized C++, C#, VB.NET, and T-SQL for SSIS and SSRS programming.
- Designed data warehouse architecture, including hardware infrastructure, schema definitions, and Fact and Relational Dimensions tables.
- Developed data warehouse and business reporting solutions to meet stakeholder requirements using SSIS, SSAS, and SSRS.

- Used SSIS ETL tools to extract and transform complex data, loading it into the data warehouse to produce Analysis Services Cubes for business reporting via SSRS or Data Marts to serve business applications.
- Created and deployed reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio, including debugging, monitoring, and troubleshooting BI solutions.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Managed and maintained company databases, ensuring data accuracy and integrity.
- Defined the target warehouse module, and created target user, module, and schema.
- Built and deployed machine learning models using k-nearest neighbors (k-NN) for customer churn prediction and fraud detection, achieving a 15% improvement in accuracy.
- Involved in UAT and TEST in a QA development environment by writing VBScript and batch files to automate processes.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI (ChatGPT-4o, Multi-GPTs, Auto-Agents), Amazon Bedrock (Claude/Titan) & SageMaker, Azure OpenAI Service, Azure ML, Azure AI Studio, Copilot, Azure AI Foundry, Google Vertex AI, Gemini API (formerly MakerSuite) , Agent-2-Agent
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
- **Workflow & Automation Platforms**: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve

PS: Interview asked OpenResearch etc.

- No, an AWS S3 bucket is not automatically a vector database.
- It is a cost-effective object storage service.1 However, AWS
- has introduced a new feature called S3 Vectors that allows an
- S3 bucket to function as a vector store for
- Retrieval-Augmented Generation (RAG) applications.2
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- When you are in the AWS Bedrock console creating a knowledge
- base, you can select Amazon S3 Vectors as the vector store
- option.3 This is a specific type of S3 bucket with built-in
- native vector storage and indexing capabilities.4
-

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-
- This feature simplifies the RAG workflow by eliminating the
- need to set up and manage a separate, more complex vector
- database like OpenSearch or Pinecone.5 It handles the
- storage, indexing, and querying of vector data directly
- within S3, while still leveraging S3's durability and

cost-effectiveness.6

When building a semantic search solution for video files on AWS, you'll need a multimodal embedding model that can process both the visual and audio components of a video and an AWS-native vector store.
When building a semantic search solution for video files on AWS, you'll need a multimodal embedding model that can process both the visual and audio components of a video and an AWS-native vector store.

Multimodal Models

These are not traditional LLMs, but they are often used in conjunction with them for tasks like semantic search. They can transform various data types, including video, into a unified vector space.

- **TwelveLabs Marengo:** This is a video-native multimodal embedding model available on Amazon Bedrock. It unifies all modalities (video, text, images, and audio) into a single vector space. It is designed for searching, classifying, summarizing, and extracting insights from video content.
- **Google's Multimodal Embedding Model:** Available through the Vertex AI platform, this model can generate embeddings for video segments. While it's a Google product, the resulting embeddings can be stored in an AWS vector database.
- **Video-LLaMA:** A multi-modal framework that integrates visual and auditory understanding into Large Language Models (LLMs) to understand video content. It uses pre-trained visual and audio encoders to create a unified embedding space.
- **Amazon Titan Multimodal Embeddings:** This model, available through Amazon Bedrock, can be used to generate embeddings from video frames and other modalities. A common approach is to extract keyframes from a video and then use this model to generate embeddings for each frame.

AWS Vector Stores

Once you've generated embeddings from your video files, you'll need a vector store to index and search them efficiently.

- **Amazon OpenSearch Service:** This is a fully managed service that provides a vector database. It's a popular choice for semantic search, offering high-performance search across billions of vectors using algorithms like Hierarchical Navigable Small World (HNSW). It integrates seamlessly with other AWS services.
- **Amazon S3 Vectors:** A new feature that allows you to store and query vector data natively within an S3 bucket. It's designed for cost-optimized and durable storage of large vector datasets, making it ideal for workloads with infrequent queries.

You can use the AWS S3 Vectors feature to semantically search a media lake. This video explains how to store and query vector data at scale in S3.

DevOps Engineer – Jenkins-EKS – Nuage – Infosys – Westpac – 5/9/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Sr. DevOps Engineer /Software Engineer /Sr. Automation Engineer/Sr. Data Engineer/Snr Network Engineer /Java/.Net Python /AI/ML/BI/MuleSoft/BizTalk/SharePoint/D365 Developer/EDW –Japan (7.5yrs) & Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), **Sydney**
Gov. Health: Sr. DevOps Engineer/ Automation Engineer/ Sr. Data Engineer/ Python/AI/ML/SP/D365 Developer/EDW
Responsibilities:

- **DevOps, Automation, & CI/CD Architecture**

- Pioneered the setup and management of a comprehensive DevOps ecosystem, leveraging Terraform to provision new AWS environments. Engineered robust, multi-stage CI/CD pipelines with Jenkins, automating the entire workflow from build to deployment using extensive Python and Bash scripting.
- Manages Jenkins agents and develops robust, multi-stage CI/CD pipelines, providing expert guidance to application teams to accelerate software delivery.
- Built automated Jenkins pipelines leveraging Jenkinsfile Runner via GitHub Actions to support comprehensive automated testing, robust deployment, and infrastructure updates in multi-environment hybrid cloud setups.
  E.g., gh workflow run 'ci-cd-pipeline.yml' --ref main
- Developed robust Python scripts and API integrations to automate interactions between infrastructure, databases, and third-party platforms, improving operational efficiency and reducing manual intervention.
- Proactively troubleshot and resolved complex automation issues across Linux-based systems, ensuring high availability and performance of automation pipelines. Delivered reusable, modular automation frameworks that integrated with diverse systems and tools via REST APIs, enabling seamless orchestration across hybrid infrastructure layers.
  E.g., dmesg -T | grep -Ei 'error|fail|warn|critical|oom'
- Pioneered the use of Large Language Models (LLMs) as optimizers within CI/CD pipelines to dynamically generate, test, and refactor code, leading to a 25% reduction in deployment time and a significant improvement in code quality across the team. E.g., llm_output = bedrock_client.invoke_model(modelId='anthropic.claude-v3', or 'amazon.titan-text-express-v1'
  body={ "prompt": "Refactor this code to optimize performance: {code_snippet}"
- Developed scalable Python applications, leveraging Generative AI frameworks (LangChain for LLM orchestration, LlamaIndex for RAG) and speech-to-text models like Whisper to automate the transcription, summarization, and analysis of customer calls. Optimized deployment of these systems on AWS (using Bedrock and OpenAI frameworks) while integrating with core business systems via data-intensive APIs (e.g., Transactions API, Balances API) and enterprise data platforms.
- Designed and implemented end-to-end automation solutions using Python and Ansible (Tower, AWX, Red Hat AAP 2.5), developing custom modules and plugins to streamline infrastructure provisioning and configuration across hybrid cloud, on-prem, and Red Hat Enterprise Linux (RHEL9/10) environments.
  E.g., ansible-playbook provision_rhel_app_servers.yml
- Automated infrastructure and application deployments using Jenkins and Terraform in Kubernetes/OpenShift clusters, ensuring scalable and repeatable workflows across cloud-agnostic platforms.
- E.g., sh 'terraform apply -auto-approve && jenkins-cli build deploy_app_on_openshift'
- Architected and deployed end-to-end AI systems from experimentation to production, applying practical ML techniques such as RAG, fine-tuning, NLP, vectorisation, and KNN to solve real-world business use cases.
  E.g., model = PeftModel.from_pretrained(model, peft_model_id).
- Automated the creation of secure, scalable Infrastructure-as-Code (IaC) CI/CD pipelines, built on secure, isolated . Generative AI platforms utilizing LLMs & self-hosted models deployed on OLLAMA4.0 within secure private virtual networks), MCP+RAG architectures, and n8n workflows. E.g., curl -X POST https://n8n.yourdomain.com/webhook/generate-cicd-pipeline
- Engineered and optimized end-to-end Retrieval-Augmented Generation (RAG) pipelines, including advanced Graph RAG and intelligent chunking with LangChain, to enhance contextual understanding and answer complex semantic queries, boosting performance and reducing API calls by over 20%.
  E.g., text_splitter = RecursiveCharacterTextSplitter(chunk_size=1000, chunk_overlap=200).
  E.g. 2: async def query_llm_with_rag(index: VectorStoreIndex, question: str) -> Dict[str, Any]
- Developed and implemented an advanced Graph-R1 agentic RAG framework, integrating structured knowledge graphs and reinforcement learning to enhance multi-hop reasoning and achieve a 25% improvement in the accuracy of semantic query responses for a knowledge-intensive platform.
- **Infrastructure as Code (IaC) & Cloud Provisioning**
- Engineered and deployed robust, automated infrastructure solutions by provisioning and configuring cloud resources with Terraform, while leveraging Vault Enterprise for centralized secrets management and secure credential distribution within CI/CD pipelines. E.g., data "vault_generic_secret" "db_creds" { path = "secret/database/credentials" }
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with AWS CloudWatch.
- **Data Processing & Testing**
- Engineered a data validation pipeline for complex projects that generates test data via API and Kafka calls, stores it on S3, and retrieves it using Athena to automatically compare results against original source data, ensuring API functionality and data integrity across multiple test scenarios.
  E.g.,sh 'aws athena start-query-execution --query-string "SELECT count(*) FROM my_test_table WHERE status = 'success'" --query-execution-context Database=your_database_name --result-configuration OutputLocation=s3://your-s3-bucket/athena-results/'
- Engineered and maintained robust ETL pipelines using AWS Glue to automate and streamline data processing, transforming and preparing data for downstream consumption.
  E.g., sh 'aws glue start-job-run --job-name my-business-data-etl-job'
  df.write.format("parquet").mode("overwrite").save("s3://processed-data-bucket/output/")

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final_df = spark.read.json("s3://raw-data/").filter("status = 'active'").dropDuplicates()

- **Big Data & Distributed Systems**
- Architected and managed Confluent Kafka infrastructure as code using Terraform, and engineered and maintained a comprehensive Terraform module library to standardize and automate the provisioning of all Kafka resources.
- Designed agent-to-agent and n8n-based orchestration workflows to streamline multi-cloud DevOps processes, integrating advanced Generative AI tools to automate the build and deployment of Next.js solutions with server-side rendering and modular architecture, ensuring resilience, compliance, and performance across aged care financial management systems.
- Professionally leveraging AWS services (S3 for scalable storage, EC2 for compute, and managed container services like EKS and ECS Fargate for scalable application deployment). Designed and implemented robust big data processing workflows and scalable ETL/ELT data pipelines using Apache Airflow and Cloud Native Distribution Infrastructure (CNDI). These pipelines incorporate advanced analytics tools such as Spark, Presto, and Hive to derive actionable insights.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov. Central Bank: Sr. DevOps Engineer/ Automation Engineer/ Sr. Data Engineer/Java/Python/AI/ML/SP/D365/BI Developer
Responsibilities:

- **DevOps, Automation, & CI/CD Architecture**
- Engineered a robust DevOps ecosystem to support mission-critical financial trading systems, leveraging Terraform to provision secure, low-latency cloud infrastructure.
- Designed and managed complex, multi-stage CI/CD pipelines with Jenkins, automating the entire software delivery lifecycle from build to deployment using extensive Python and Bash scripting.
- Manages Jenkins agents and develops robust, multi-stage CI/CD pipelines, providing expert guidance to application teams to accelerate software delivery.
- Automated resilient infrastructure and real-time application deployments using Jenkins and Terraform within Kubernetes/OpenShift clusters, ensuring scalable and repeatable workflows for mission-critical financial trading services across cloud-agnostic platforms.
- E.g., sh 'terraform apply -auto-approve && jenkins-cli build deploy_low_latency_trading_engine'
- Specialized in designing and implementing end-to-end automation solutions using Python and Red Hat Ansible Automation Platform (including Ansible Tower, and Ansible Web eXecutor(AWX), developing custom modules and plugins to streamline secure, low-latency infrastructure provisioning and configuration for RBA financial trading platforms across hybrid cloud, on-prem, and RHEL9 environments. E.g., ansible-playbook apply_rba_security_baseline.yml
- **Rule-based & Self-Learning Systems**
- Developed self-learning Multi-agent systems (MAS), a subfield of distributed AI in Python, to automate and optimize algorithmic strategies, enabling models to autonomously adapt to dynamic financial market patterns.
- Engineered intelligent rule-based Robotic Process Automation (RPA) workflows powered by NLP libraries (e.g., NLTK, spaCy) for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Developed scalable Python applications, leveraging frameworks like scikit-learn and NLTK for data indexing and natural language processing. E.g., model.predict(vectorizer.transform(["RBA interest rate hike triggers AUD volatility"]))
- **Infrastructure as Code (IaC) & Cloud Provisioning**
- Engineered and deployed robust, automated infrastructure solutions for a financial trading environment. This included provisioning and configuring secure cloud resources with Terraform, while leveraging Vault Enterprise for centralized secrets management and secure credential distribution within CI/CD pipelines.
  E.g., data "vault_generic_secret" "db_creds" { path = "secret/database/credentials" }
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with AWS CloudWatch to ensure regulatory compliance and operational transparency.
- Automated resilient infrastructure and real-time application deployments using Ansible and Terraform within Kubernetes/OpenShift clusters, ensuring scalable and repeatable workflows for mission-critical financial trading services across cloud-agnostic platforms.
- E.g., terraform apply -auto-approve && ansible-playbook deploy_low_latency_trading_engine.yml
- Specialized in designing and implementing end-to-end automation solutions using Python and Red Hat Ansible Automation Platform, developing custom modules and plugins to streamline secure, low-latency infrastructure provisioning and configuration for RBA financial trading platforms across hybrid cloud, on-prem, and RHEL9 environments.
  E.g., ansible-playbook apply_rba_security_baseline.yml
- **Big Data & Distributed Systems**
- Architected and managed a high-throughput Confluent Kafka infrastructure using Terraform, engineering a comprehensive module library to standardize the automated provisioning of all Kafka topics and services. This framework ensures reliable, low-latency data streaming for real-time market data and financial transactions.
- E.g., resource "confluent_kafka_topic" "market_data_stream" { ... }
- Developed robust, real-time big data processing workflows and scalable ETL/ELT data pipelines using Apache Airflow. These pipelines incorporate advanced analytics tools such as Spark, Presto, and Hive to derive actionable insights from market and transactional data.

- E.g., df = spark.read.parquet("s3a://raw-trade-data/").filter(col("status") == "cleared")
- **Data Processing & Testing**
- Architected and maintained a robust data validation and processing framework crucial for financial trading systems. Engineered a specialized data validation pipeline that generates test data via API and Kafka calls, stores it securely on S3, and retrieves it using Athena to automatically compare results against original source data, ensuring the integrity and accuracy of all trade-related data.
- E.g., sh 'aws athena start-query-execution --query-string "SELECT count(*) FROM my_trading_transactions WHERE status = 'cleared'"
- Engineered and maintained robust ETL pipelines using AWS Glue to automate and streamline data processing, transforming and preparing raw trade data for downstream consumption and analysis.
- E.g., sh 'aws glue start-job-run --job-name rba-trading-data-etl-job'

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Sr. DevOps Engineer/ Automation Engineer/Sr. Data Engineer/Python/ML/SharePoint/D365 Developer
Responsibilities:

- **DevOps, Automation, & CI/CD Architecture**
- Engineered a robust DevOps ecosystem to support mission-critical university financial systems, leveraging Terraform to provision secure, high-availability cloud infrastructure.
- Designed and managed complex, multi-stage CI/CD pipelines with Jenkins, automating the entire software delivery lifecycle for student enrollment and financial aid applications using extensive Python and Bash scripting.
- Developed and maintained Jenkins agents and multi-stage CI/CD pipelines, providing expert guidance to application teams to accelerate secure, compliant software delivery.
- Automated resilient infrastructure and real-time application deployments using Jenkins and Terraform within Kubernetes/OpenShift clusters, ensuring scalable and repeatable workflows for mission-critical university financial and enrollment systems across cloud-agnostic platforms.
- E.g., sh 'terraform apply -auto-approve && jenkins-cli build deploy_enrollment_engine'
- Specialized in designing and implementing end-to-end automation solutions using Python and Red Hat Ansible Automation Platform, developing custom modules and plugins to streamline secure infrastructure provisioning and configuration for university platforms across hybrid cloud, on-prem, and RHEL9 environments.
- E.g., ansible-playbook apply_university_security_baseline.yml
- **Rule-based & Self-Learning Systems**
- Developed self-learning Multi-agent systems (MAS), a subfield of distributed AI in Python, to automate and optimize algorithmic strategies, enabling models to autonomously adapt to dynamic student enrollment patterns and financial aid distribution.
- Engineered intelligent rule-based Robotic Process Automation (RPA) workflows powered by NLP libraries (e.g., NLTK, spaCy) for automated student inquiry qualification and interaction analysis, improving data-driven decision-making in the financial aid department.
- Developed scalable Python applications, leveraging frameworks like scikit-learn and NLTK for data indexing and natural language processing. E.g., model.predict(vectorizer.transform(["student loan application submitted"]))
- **Infrastructure as Code (IaC) & Cloud Provisioning**
- Engineered and deployed robust, automated infrastructure solutions for a university financial environment. This included provisioning and configuring secure cloud resources with Terraform, while leveraging Vault Enterprise for centralized secrets management and secure credential distribution within CI/CD pipelines.
- E.g., data "vault_generic_secret" "api_creds" { path = "secret/university/api/credentials" }
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with AWS CloudWatch to ensure regulatory compliance and operational transparency for financial and student data.
- Automated resilient infrastructure and real-time application deployments using Ansible and Terraform within Kubernetes/OpenShift clusters, ensuring scalable and repeatable workflows for mission-critical university systems across cloud-agnostic platforms. E.g., terraform apply -auto-approve && ansible-playbook deploy_student_portal.yml
- Specialized in designing and implementing end-to-end automation solutions using Python and Red Hat Ansible Automation Platform, developing custom modules and plugins to streamline secure infrastructure provisioning and configuration for university platforms across hybrid cloud, on-prem, and RHEL9 environments.
- E.g., ansible-playbook apply_university_security_baseline.yml
- **Big Data & Distributed Systems**
- Architected and managed a high-throughput Confluent Kafka infrastructure using Terraform, engineering a comprehensive module library to standardize the automated provisioning of all Kafka topics and services. This framework ensures reliable data streaming for real-time student enrollment data and financial transactions.
- E.g., resource "confluent_kafka_topic" "student_data_stream" { ... }

- Developed robust, real-time big data processing workflows and scalable ETL/ELT data pipelines using Apache Airflow. These pipelines incorporate advanced analytics tools such as Spark, Presto, and Hive to derive actionable insights from student and transactional data. E.g., df = spark.read.parquet("s3a://raw-enrollment-data/").filter(col("status") == "approved")
- **Data Processing & Testing**
- Architected and maintained a robust data validation and processing framework crucial for the university's financial and enrollment systems. Engineered a specialized data validation pipeline that generates test data via API and Kafka calls, stores it securely on S3, and retrieves it using Athena to automatically compare results against original source data, ensuring the integrity and accuracy of all student and financial data. E.g., sh 'aws athena start-query-execution --query-string "SELECT count(*) FROM my_enrollment_transactions WHERE status = 'cleared'"
- Engineered and maintained robust ETL pipelines using AWS Glue to automate and streamline data processing, transforming and preparing raw student data for downstream consumption and analysis.
- E.g., sh 'aws glue start-job-run --job-name university-data-etl-job'

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: DevOps Engineer/ ML Engineer /Automation Engineer/Software Test Engineer / Data Engineer /EDW
Responsibilities:

- **Application Development and Machine Learning**
- Developed self-learning multi-agent systems in Python to automate and optimize the processing of injury claims, enabling autonomous adaptation to claims data and improving overall system efficiency.
- Python Application Developer with a robust quantitative foundation, adept at building high-performance solutions. Proficient in Python and Java, with capabilities to design, develop, and optimize ML applications and deliver actionable insights.
- Developed and deployed predictive models using Python libraries like scikit-learn to accurately assess the likelihood of specific claim outcomes or potential fraud indicators, enabling proactive risk management and optimized resource allocation in the injury compensation domain.
- Back-End Development: Engineered secure and scalable backend services using Python (e.g., Flask) and Java to efficiently manage high volumes of injury compensation claim data, ensuring data integrity and seamless integration with critical EML systems.
- Front-End Development: Designed and implemented user interfaces with jQuery and JavaScript, optimizing the claimant experience for EML's injury compensation portal and streamlining data presentation for complex claim workflows.
- Extensive expertise in programming languages such as Java, Python, C#, and PowerShell scripting languages.
- **Cloud & Data Engineering**
- Architected and implemented data solutions on AWS, focusing on traditional data warehousing and ETL processes.
- **Batch laye**r: Utilized Amazon EMR, a managed service for running Apache Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed AWS Redshift for near-real-time data analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Designed and developed data solutions using AWS services like EC2, RDS, and S3.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created mapping documents that defined how source data fields corresponded to target data fields.
- Implemented mapping logic based on the mapping document, including writing code to map the source data to the target data according to defined rules.
- **DevOps & Automation**
- Automated deployment of scalable machine learning applications and services, leveraging AWS services like EC2 to ensure high availability.
- Championed continuous integration and delivery (CI/CD) pipelines using tools like Jenkins for secure, continuous application delivery.
- Utilized PowerShell scripting to automate system configurations, testing, and deployment processes.
- Followed software development best practices for versioning, deployment, testing, and automation.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer /Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Software Development and Deployment: Architected and built scalable web applications using Python and deployed them on traditional servers.
- Infrastructure Management: Managed infrastructure using scripting languages and automation tools.
- Application Development: Developed applications using Python for data processing and business logic.
- Data Management: Utilized relational databases for data storage and retrieval.
- System Integration: Implemented APIs and integrated with diverse systems.

- Automated build and deployment processes using Jenkins and shell scripting, managing server configurations with Puppet to ensure reliable delivery of machine learning models for research projects.
- Deployed and maintained web applications on virtual servers. Managed infrastructure using shell scripts and other automation tools, integrating them into CI/CD pipelines (e.g., Jenkins). Developed REST APIs using Python and Java.
- Monitored server resources using on-premise tools and implemented logging, monitoring, and security measures.
- **Security and Compliance**: Implemented security best practices and regularly patched systems and applications.
- **Agile and Data Expertise**: Experienced working in agile environments with strong data engineering practices.
- Other Proficiencies: Possessed a strong understanding of SQL, Python scripting, and SharePoint.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of continuous integration (CI) and automated deployment processes, leading to a significant reduction in software feature delivery time.
- Utilized scripting languages and configuration management tools to implement Infrastructure as Code (IaC) principles, enhancing deployment processes and scalability.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Administered and monitored Unix/Linux-based systems, ensuring the smooth operation of critical applications.
- Provided Level 2/3 technical support for all Microsoft infrastructures, including Active Directory (AD), DNS, DHCP, Exchange 2010, and SharePoint.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running costs.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Served as Project Manager to replace the existing insurance web application (from IBM AS/400 and LAMP) with a WISA-based system, significantly reducing IT operational costs and integrating with a Salesforce Automation (SFA) platform for improved marketing results.
- Planned, implemented, and supported all core IT systems including Hyper-V, VMware ESX & vSphere 4.x, HP/EMC/Dell SAN Storage, and Windows Server 2008, and applied security principles and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for desktop productivity suite issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Sr. Systems Engineer /EDW
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for Business Intelligence (BI) solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized C++, C#, VB.NET, and T-SQL for SSIS and SSRS programming.
- Designed data warehouse architecture, including hardware infrastructure, schema definitions, and Fact and Relational Dimensions tables.
- Developed data warehouse and business reporting solutions to meet stakeholder requirements using SSIS, SSAS, and SSRS.
- Used SSIS ETL tools to extract and transform complex data, loading it into the data warehouse to produce Analysis Services Cubes for business reporting via SSRS or Data Marts to serve business applications.
- Created and deployed reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio, including debugging, monitoring, and troubleshooting BI solutions.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Managed and maintained company databases, ensuring data accuracy and integrity.
- Defined the target warehouse module, and created target user, module, and schema.
- Built and deployed machine learning models using k-nearest neighbors (k-NN) for customer churn prediction and fraud detection, achieving a 15% improvement in accuracy.
- Involved in UAT and TEST in a QA development environment by writing VBScript and batch files to automate processes.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI (ChatGPT-4o, Multi-GPTs, Auto-Agents), Amazon Bedrock (Claude/Titan) & SageMaker, Azure OpenAI Service, Azure ML, Azure AI Studio, Copilot, Azure AI Foundry, Google Vertex AI, Gemini API (formerly MakerSuite) , Agent-2-Agent
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
- **Workflow & Automation Platforms**: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve

Sr. AI Engineer– Bedrock-SageMaker Specialist – CBA – Opus Peter – 2/9/2025 – F.it No resp and advertised next day
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior AI Engineer– Bedrock-SageMaker Specialist / GenAI | AI Build | Automation Engineer/ Snr DevOps /Data Engineer
Principle Software Engineer/Full Stack Developer(Py/Java/.Net/Mulesoft /AI/ML/DL/BI/SP/D365)-Japan (7.5yrs) & Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior AI Engineer–Bedrock-SageMaker Specialist/ GenAI | Principle Software Engineer | Automation Engineer
Responsibilities:

- **Agentic Platform Development & Generative AI**
- Architected and deployed end-to-end AI systems from experimentation to production, applying practical ML techniques such as RAG, fine-tuning, NLP, vectorisation, and KNN to solve real-world business use cases.
  E.g., model = PeftModel.from_pretrained(model, peft_model_id).
- Engineered and optimized end-to-end Retrieval-Augmented Generation (RAG) pipelines, including advanced Graph RAG and intelligent chunking with LangChain, to enhance contextual understanding and answer complex semantic queries, boosting performance and reducing API calls by over 20%.
  E.g., text_splitter = RecursiveCharacterTextSplitter(chunk_size=1000, chunk_overlap=200).
  E.g. 2: async def query_llm_with_rag(index: VectorStoreIndex, question: str) -> Dict[str, Any]:
- Developed and implemented an advanced Graph-R1 agentic RAG framework, integrating structured knowledge graphs and reinforcement learning to enhance multi-hop reasoning and achieve a 25% improvement in the accuracy of semantic query responses for a knowledge-intensive platform.

- Developed and orchestrated sophisticated autonomous AI agents using frameworks like LangGraph, AutoGen, and Semantic Kernel, and leveraging inter-agent communication protocols and tools like n8n to integrate with external APIs, driving operational efficiency and intelligent decision-making.
  E.g, response = await kernel.invoke_async("email_plugin.summarize", text=email_body).
- Architected and deployed enterprise-grade Generative AI solutions using Amazon Bedrock, leveraging a suite of pre-trained foundation models including Anthropic's Claude 3 for its long context window and Amazon Titan. Applied prompt engineering and LLM-based RAG to build a context-aware customer support chatbot, reducing response time by 30%.
- Utilized cutting-edge Graph-R1 and Reinforcement Learning to build a robust Agentic Graph RAG framework, successfully reducing hallucination rates by configuring security-level parameters.
- Developed a comprehensive observability framework for data platforms, leveraging insights from CloudWatch Logs to predict ETL failures in Airflow and Glue jobs and implementing automated remediation patterns to reduce incident response time by over 40%. E.g., boto3.client('lambda').invoke(FunctionName='AutomatedRemediationFunction', InvocationType='Event', Payload='{"failure_type": "ETL_Error", "job_id": "job_123"}')
- Consistently delivered high-quality AI solutions within agile sprint cycles, using data-driven metrics like MTTR and FPR to evaluate performance and iterate on improvements, which allowed for effective communication of complex AI logic to non-technical stakeholders via dashboards and chat interfaces.
- **AI-Powered Automation & Insights**
- Engineered intelligent RPA workflows powered by NLP for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Developed scalable Python applications, leveraging Generative AI frameworks (LangChain for LLM orchestration, LlamaIndex for RAG) and speech-to-text models like Whisper to automate the transcription, summarization, and analysis of customer calls. Optimized deployment of these systems on AWS (using Bedrock and OpenAI frameworks) while integrating with core business systems via data-intensive APIs (e.g., Transactions API, Balances API) and enterprise data platforms.
- Integrated MongoDB Atlas as a vector database for RAG (Retrieval-Augmented Generation), enabling real-time and context-aware enterprise insights.
- Developed an AI-powered insights platform for a major financial institution by creating a knowledge graph to map complex customer and transaction data. Used embeddings to represent this information as high-dimensional vectors, which were then stored in a vector database (Pinecone/Weaviate) to enable lightning-fast semantic search and relationship discovery, enhancing fraud detection and personalized financial recommendations.
- Developed an intelligent AI agent using an Amazon Scraper (Python, requests, BeautifulSoup, pandas) with Streamlit UI, demonstrating robust data acquisition and constraints bypassing.
- **Machine Learning & Model Development**
- Developed, trained, and deployed a suite of machine learning and deep learning models on Amazon SageMaker using PyTorch, TensorFlow, and Hugging Face Transformers, optimizing data pipelines and automating model retraining to improve predictive accuracy and deliver NLP solutions.
- Built robust predictive models and secure AI APIs using diverse machine learning algorithms (KNN, CNN, RNN, ANN, GAN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning), applying Scikit-learn and Keras to drive Next Best Action (NBA) recommendations and pattern discovery.
- Drove the full lifecycle of AI solutions, from ideation and opportunity identification with key stakeholders to the design and delivery of scalable prototypes. Conducted comprehensive experiments to evaluate model performance, iterating on design to optimize outcomes and ensure compliance with ethical AI standards.

- **DevOps, Cloud Infrastructure & Backend Development**
- Pioneered the use of LLMs and agentic workflows to automate and refactor code within scalable CI/CD pipelines, reducing deployment time by 25%. Engineered and deployed mission-critical Python applications and GenAI platforms on a multi-cloud architecture (AWS, Azure, GCP), leveraging IaC, MCP+RAG, and n8n orchestration.
  - E.g., llm_output = gemini.generate_text(f"Refactor this code to optimize performance: {code_snippet}")
  - E.g., ansible-playbook provision_rhel_app_servers.yml
  - E.g., requests.post(f"{api_url}/api/v1/deployments", json={'app':'genai-service', 'version':'v1.0.0'}).
- Leveraged AWS Kiro's spec-driven development & agentic capabilities to design, build, and maintain scalable, secure AWS infrastructure (EC2, EKS, S3, RDS, Lambda, SageMaker, Amazon Neptune) for hosting advanced Graph RAG pipelines, vector databases, & LLMs, Using AWS Bedrock for diverse foundational models (e.g., Amazon, Anthropic, AI21 Labs).
- Implemented cloud-native architectures with AWS ECS/EKS for containerized microservices and REST APIs, ensuring efficient model integration with enterprise platforms streamlined data ingestion within an AWS Redshift-powered lakehouse.
- Designed and developed scalable backend APIs and seamless integrations using Python frameworks (Django, Flask), optimized data pipelines for AI model inference, enabling real-time predictions on large datasets.
- Pioneered the use of Large Language Models (LLMs) as optimizers within CI/CD pipelines to dynamically generate, test, and refactor code, leading to a 25% reduction in deployment time and a significant improvement in code quality across the team. E.g., llm_output = bedrock_client.invoke_model(modelId='anthropic.claude-v3', or 'amazon.titan-text-express-v1'
  body={ "prompt": "Refactor this code to optimize performance: {code_snippet}"
- **Technical Leadership & Innovation**

Zhi Ming ZHANG - Australian Citizen
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Mobile: 0402-997-679 | mingjp@gmail.com

- Led end-to-end GenAI solution development from secure audio ingestion, automated transcription, LLM orchestration, RAG pipeline integration, to intelligent summarization—ensuring seamless data flow across multi-cloud.
- Leveraged a self-recursive Artificial Superintelligence Architecture (ASI-ARCH) on the Harrison Platform (Nvidia DGX-1 systems) to autonomously innovate and optimize AI models for financial management. Enhanced algorithmic trading strategies and risk mitigation by discovering emergent market principles beyond human intuition.
- Integrated M365 environments (O365, MS Teams, SharePoint online, Dynamics 365, Power Automate,Power App, Power BI) & AI Builder for D365 into the AWS Landing Zones for a comprehensive and secure platform for Microsoft workloads.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov Central Bank: Senior AI/ML Engineer/Automation Engineer/Snr DevOps/Data Engineer /AI/ML/DL/Mulesoft Developer
Responsibilities:

- AI/ML-Powered Automation & Insights
- Developed self-learning Multi-agent systems (MAS), a subfield of distributed AI in Python to automate and optimize algorithmic strategies, enabling models to autonomously adapt to dynamic financial market patterns.
- Engineered intelligent rule-based Robotic Process Automation (RPA) workflows powered by NLP libraries (e.g., NLTK, spaCy) for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Developed scalable Python applications, leveraging frameworks like scikit-learn and NLTK for data indexing and natural language processing.E.g., model.predict(vectorizer.transform(["RBA interest rate hike triggers AUD volatility"]))
- Led end-to-end machine learning solution development from secure audio ingestion, automated transcription, to intelligent summarization—ensuring seamless data flow across multi-cloud.Integrated speech-to-text APIs for automatic high-volume call transcription and sentiment analysis, directly logging customer sentiment into CRM workflows.
- Developed computer vision models using OpenCV & Scikit-image to integrate image recognition for automated validation of property documents & anomaly detection, streamlining regulatory workflows & enhancing decision-making accuracy.
- Technical Foundations & Modeling
- Built robust predictive models and secure machine learning APIs using diverse algorithms (KNN, CNN, RNN, ANN, GAN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning), applying Scikit-learn and Keras to drive next best action (NBA) recommendations and pattern discovery.
- Leveraged PyTorch and TensorFlow to develop and deploy deep learning models, including fine-tuning pre-trained models for NLP and analytics.Utilized Nvidia DGX-1 systems to develop and optimize machine learning models for financial management.
- Enhanced algorithmic trading strategies and risk mitigation by discovering new market principles beyond human intuition.Engineered scalable, mission-critical production applications in Python, driving robust system integrations and continuous deployment for machine learning services with core code such as requests.post(f"{api_url}/api/v1/deployments", json={'app':'ml-service', 'version':'v1.0.0'}).
- DevOps & Cloud Infrastructure
- Automated the creation of secure, scalable IaC CI/CD pipelines and isolated machine learning platforms, enabling robust deployment of Python (Django, Flask) and other server-side applications across AWS, Azure, and GCP.
- Implemented cloud-native architectures with AWS ECS for containerized microservices and REST APIs, ensuring efficient model integration with enterprise platforms and streamlined data ingestion within a data lakehouse.
- Specialized in designing and implementing end-to-end automation solutions using Python & Red Hat Ansible Automation Platform (including Ansible Tower and AWX), developing modules and plugins to streamline secure, low-latency infrastructure provisioning for RBA trading platforms across hybrid cloud, on-prem, and RHEL9 environments. E.g., ansible-playbook deploy_low_latency_trading_engine.yml to automate trading engine deployments.
- Leveraged PagerDuty as a central incident management system, effectively aggregating and transforming alerts from diverse monitoring and observability platforms (e.g., Airflow and Glue ETL failures, Datadog, Prometheus, AWS services) into actionable incidents.
- Integrated PagerDuty with Amazon DevOps Guru to leverage its AIOps/ML-powered operational insights, translating them into actionable incidents, triggering a PagerDuty incident. Runbook Automation via Bamboo: Reduce support costs by 50% by eliminating toil with PagerDuty. E.g., aws cloudwatch put-metric-data --metric-name CriticalApplicationError --namespace MyAIOpsApplication --value 1 --unit Count --dimensions Environment=Production,Service=AuthService.
- Developed and deployed scalable applications using MuleSoft Anypoint Platform, including Mule 4 & Anypoint Studio 7.x, with a strong focus on robust backend API development and seamless system integrations.E.g. 1: POST /api/v1/orders HTTP/1.1 { "item": "widget", "qty": 2 }E.g. 2: Flow orchestrates Salesforce, SAP, and custom REST services for order processing.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr ML Engineer/Automation Engineer /Integration Solution Architect /ML/DL/Mulesoft Developer
Responsibilities:

- **Automated Systems and Machine Learning**

- Engineered self-learning Multi-agent systems (MAS) in Python to automate the management of student enrollment and financial data, enabling models to autonomously adapt to complex patterns for enhanced system performance.
- Engineered intelligent rule-based Robotic Process Automation (RPA) workflows powered by NLP libraries (e.g., NLTK, spaCy) for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Developed computer vision models (using OpenCV and Scikit-image) to integrate image recognition for the automated analysis of visual data and anomaly detection, streamlining data processing and enhancing decision-making accuracy.
- Led end-to-end machine learning project development, from data collection and automated transcription to statistical analysis and intelligent summarization, ensuring efficient data flow for research and academic projects.
- Developed scalable Python applications, leveraging core frameworks like scikit-learn, NLTK, Theano, and Caffe for data preprocessing, modeling, and natural language processing. Utilized third-party speech-to-text services for automatic transcription and sentiment analysis, integrating results into custom-built data analysis pipelines.
- **Machine Learning and Data Insights**
- Built robust predictive models and secure APIs using diverse machine learning algorithms (KNN, CNN, RNN, ANN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning). Applied Scikit-learn and Keras for projects focused on pattern discovery and predictive analytics.
- Utilized GPU-accelerated systems to develop and optimize machine learning models for research projects. Developed and tested algorithmic strategies by applying emerging principles in statistical analysis.Integrated databases like PostgreSQL or MongoDB for managing and analyzing data in projects, enabling real-time and context-aware insights.
- **Systems Engineering and Automation**
- Implemented cloud-based architectures with AWS ECS for containerized microservices and REST APIs, ensuring efficient model integration and streamlined data ingestion.
- Automated the creation of secure and scalable infrastructure for machine learning experimentation using Infrastructure-as-Code (Terraform) and automated CI/CD pipelines (e.g., Jenkins) for robust deployment across cloud platforms like AWS, Azure, and GCP.
- Used AWS infrastructure (EC2, S3, and RDS) to design, build, and maintain scalable systems for hosting ML models.
- Designed and developed scalable backend APIs and integrations using Python frameworks (Django, Flask), optimizing data pipelines for efficient machine learning model training and enabling near real-time predictions on datasets.
- Developed an intelligent web scraper (Python, requests, BeautifulSoup, pandas) with a simple UI built using Django or Flask to demonstrate robust data acquisition and processing capabilities.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: ML Engineer /Automation Engineer/Software Test Engineer /DevOps/Data Engineer
Responsibilities:

- **Application Development and Machine Learning**
- Developed self-learning multi-agent systems in Python to automate and optimize the processing of injury claims, enabling autonomous adaptation to claims data and improving overall system efficiency.
- Developed and deployed predictive models using Python libraries like scikit-learn to accurately assess the likelihood of specific claim outcomes or potential fraud indicators, enabling proactive risk management and optimized resource allocation in the injury compensation domain.
- Python Application Developer with a robust quantitative foundation, adept at building high-performance solutions. Proficient in Python and Java, with capabilities to design, develop, and optimize ML applications and deliver actionable insights.
- Engineered secure and scalable backend services using Python (e.g., Flask) and Java to efficiently manage high volumes of injury compensation claim data, ensuring data integrity and seamless integration with critical EML systems.
- Designed and implemented user interfaces with jQuery and JavaScript, optimizing the claimant experience for EML's injury compensation portal and streamlining data presentation for complex claim workflows.
- Extensive expertise in programming languages such as Java, Python, C#, and PowerShell scripting languages.Data Engineering & ArchitectureArchitected and implemented data solutions on AWS, focusing on traditional data warehousing and ETL processes.
- Utilized Amazon EMR for batch processing of large data volumes, and AWS Redshift for near-real-time data analysis.
- Employed Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Designed and developed data solutions using AWS services like EC2, RDS, and S3.Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created and implemented mapping documents that defined how source data fields corresponded to target data fields, including writing code to map the data according to defined rules.
- DevOps & Automation

- Automated deployment of scalable machine learning applications and services, leveraging AWS services like EC2 to ensure high availability.Championed continuous integration and delivery (CI/CD) pipelines using tools like Jenkins for secure, continuous application delivery.
- Utilized PowerShell scripting to automate system configurations, testing, and deployment processes.Followed software development best practices for versioning, deployment, testing, and automation.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps/Data Engineer /Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Software Development and Deployment: Architected and built scalable web applications using Python and deployed them on traditional servers.
- Infrastructure Management: Managed infrastructure using scripting languages and automation tools.
- Application Development: Developed applications using Python for data processing and business logic.
- Data Management: Utilized relational databases for data storage and retrieval.
- System Integration: Implemented APIs and integrated with diverse systems.
- Automated build and deployment processes using Jenkins and shell scripting, managing server configurations with Puppet to ensure reliable delivery of machine learning models for research projects.
- Deployed and maintained web applications on virtual servers. Managed infrastructure using shell scripts and other automation tools, integrating them into CI/CD pipelines (e.g., Jenkins). Developed REST APIs using Python and Java.
- Monitored server resources using on-premise tools and implemented logging, monitoring, and security measures.
- **Security and Compliance**: Implemented security best practices and regularly patched systems and applications.
- **Agile and Data Expertise**: Experienced working in agile environments with strong data engineering practices.
- Other Proficiencies: Possessed a strong understanding of SQL, Python scripting, and SharePoint.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of continuous integration (CI) and automated deployment processes, leading to a significant reduction in software feature delivery time.
- Utilized scripting languages and configuration management tools to implement Infrastructure as Code (IaC) principles, enhancing deployment processes and scalability.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Administered and monitored Unix/Linux-based systems, ensuring the smooth operation of critical applications.
- Provided Level 2/3 technical support for all Microsoft infrastructures, including Active Directory (AD), DNS, DHCP, Exchange 2010, and SharePoint.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running costs.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Served as Project Manager to replace the existing insurance web application (from IBM AS/400 and LAMP) with a WISA-based system, significantly reducing IT operational costs and integrating with a Salesforce Automation (SFA) platform for improved marketing results.
- Planned, implemented, and supported all core IT systems including Hyper-V, VMware ESX & vSphere 4.x, HP/EMC/Dell SAN Storage, and Windows Server 2008, and applied security principles and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for desktop productivity suite issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for Business Intelligence (BI) solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized C++, C#, VB.NET, and T-SQL for SSIS and SSRS programming.
- Designed data warehouse architecture, including hardware infrastructure, schema definitions, and Fact and Relational Dimensions tables.
- Developed data warehouse and business reporting solutions to meet stakeholder requirements using SSIS, SSAS, and SSRS.

- Used SSIS ETL tools to extract and transform complex data, loading it into the data warehouse to produce Analysis Services Cubes for business reporting via SSRS or Data Marts to serve business applications.
- Created and deployed reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio, including debugging, monitoring, and troubleshooting BI solutions.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Managed and maintained company databases, ensuring data accuracy and integrity.
- Defined the target warehouse module, and created target user, module, and schema.
- Built and deployed machine learning models using k-nearest neighbors (k-NN) for customer churn prediction and fraud detection, achieving a 15% improvement in accuracy.
- Involved in UAT and TEST in a QA development environment by writing VBScript and batch files to automate processes.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI (ChatGPT-4o, Multi-GPTs, Auto-Agents), Amazon Bedrock (Claude/Titan) & SageMaker, Azure OpenAI Service, Azure ML, Azure AI Studio, Copilot, Azure AI Foundry, Google Vertex AI, Gemini API (formerly MakerSuite) , Agent-2-Agent
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
- **Workflow & Automation Platforms**: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve

AI Engineer – GKE, AKE - Correlated Resources – Kiara – 30/8/2025 – No resp.
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior AI Engineer-Python / GenAI | Greenfield AI Build | Enterprise Innovation /Automation Engineer/Full Stack Develop/r Software Engineer/Snr DevOps /Data Engineer /Mulesoft /AI/ML/DL/BI/SP/D365 Developer -Japan (7.5yrs) and Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior AI Engineer /GenAI | Python/AI/ML Developer | Snr DevOps/Data Engineer /Automation Engineer/EDW
Responsibilities:

- **Agentic Platform Development & Generative AI**
- Architected and deployed end-to-end AI systems from experimentation to production, applying practical ML techniques such as RAG, fine-tuning, NLP, vectorisation, and KNN to solve real-world business use cases,
  E.g., model = PeftModel.from_pretrained(model, peft_model_id).

- Built and optimized Retrieval-Augmented Generation (RAG) pipelines leveraging a combination of efficient retrievers and powerful LLMs (Gemini, Azure OpenAI) for enhanced contextual understanding and response generation. Focused on pre-retrieval optimization through intelligent chunking and metadata enrichment using LangChain to improve retrieval accuracy. E.g., text_splitter = RecursiveCharacterTextSplitter(...)
- Integrated advanced Graph RAG and implemented an optimized RAG pipeline to answer semantic queries, significantly reducing API calls and boosting performance by 20%. E.g. 1: kg_builder = SimpleKGPipeline(llm=ex_llm, driver=neo4j_driver, ...) E.g. 2: async def query_llm_with_rag(index: VectorStoreIndex, question: str) -> Dict[str, Any]:
- Utilized inter-agent communication protocols and integrated external tools (e.g., web search APIs) with n8n for enhanced operational efficiency. E.g, response = await kernel.invoke_async("email_plugin.summarize", text=email_body)
- Utilized cutting-edge Graph-R1 and Reinforcement Learning to build a robust Agentic Graph RAG framework, successfully reducing hallucination rates by configuring security-level parameters.
- Leveraged expertise in both GCP Vertex AI Agent Builder and Azure AI Foundry to architect and deploy comprehensive Agentic AI solutions. Built robust APIs and backend integrations while developing scalable deployment pipelines that brought a range of AI models—from Generative AI to LLMs—into live customer products and internal platforms.
- Developed advanced agentic applications and autonomous AI solutions using frameworks like LangGraph, AutoGen, CrewAI, and Semantic Kernel. E.g.: researcher = Agent(role="Senior Researcher", goal="Find market trends")
- Architected self-learning AI agent systems that adapt based on customer patterns, significantly increasing automation and system performance. E.g.: agent.learn_from(customer_transaction_data)
- **AI-Powered Automation and Insights**
- Developed an AI-powered insights platform for a major financial institution by creating a knowledge graph to map complex customer and transaction data. Used embeddings to represent this information as high-dimensional vectors, which were then stored in a vector database (Pinecone/Weaviate) to enable lightning-fast semantic search and relationship discovery, enhancing fraud detection and personalized financial recommendations.
- Integrated MongoDB Atlas as a vector database for RAG (Retrieval-Augmented Generation), enabling real-time and context-aware enterprise insights.
- AI-Augmented RPA with Natural Language Processing (NLP): Engineered intelligent RPA workflows powered by NLP for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Integrated speech-to-text APIs like Google Cloud Speech-to-Text for automatic high-volume call transcription and sentiment analysis, directly logging customer sentiment into CRM workflows.
- Developed scalable Python applications leveraging frameworks like LangChain and LlamaIndex with speech-to-text models like Whisper to automate customer call analysis. Optimized deployment of these systems on GCP (using Vertex AI for model management and fine-tuning) while integrating with APIs, microservices, and enterprise data platforms.
- **Machine Learning and Model Development**
- Leveraged PyTorch, TensorFlow, and Hugging Face Transformers to develop and deploy LLMs (BERT, GPT-4o, Jurassic-1, Bloom) and deep learning models, including fine-tuning pre-trained models for NLP and analytics.
- Built robust predictive models and secure AI APIs using diverse machine learning algorithms (KNN, CNN, RNN, ANN, GAN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning), applying Scikit-learn and Keras to drive Next Best Action (NBA) recommendations and pattern discovery.
- Drove the full lifecycle of AI solutions, from ideation and opportunity identification with key stakeholders to the design and delivery of scalable prototypes. Conducted comprehensive experiments to evaluate model performance, iterating on design to optimize outcomes and ensure compliance with ethical AI standards.
- **DevOps, Cloud Infrastructure & Backend Development**
- Engineered scalable and robust solutions in Python and Golang, developing APIs, backend integrations, and deployment pipelines across GCP, Azure, and Kubernetes with Docker. E.g. 1: E.g., http.HandleFunc("/api/data", getData) E.g. 2: requests.post(f"{api_url}/api/v1/deployments", json={'app':'genai-service', 'version':'v1.0.0'})
- Designed and built end-to-end production-ready AI systems, creating scalable backend integrations and deployment pipelines on GCP/Azure and Kubernetes. Applied practical ML techniques such as RAG, fine-tuning, and NLP to deliver high-impact features. Including Perception Module, Reasoning Engine and Action Agent.
- Developed and deployed scalable deployment pipelines and backend APIs to integrate AI models into live customer products and internal platforms, including backend logic, database operations, and interfaces for calling external models.
- Pioneered the use of Large Language Models (LLMs) as optimizers within CI/CD pipelines to dynamically generate, test, and refactor code, leading to a 25% reduction in deployment time and a significant improvement in code quality across the team. E.g., llm_output = gemini.generate_text(f"Refactor this code to optimize performance: {code_snippet}")
- Automated the deployment of scalable GenAI applications and microservices across hybrid cloud environments, leveraging cloud-native Google GKE, Cloud Functions, & Vertex AI to ensure high availability & streamlined CI/CD pipelines.
- Automated creation of secure, scalable IaC CI/CD pipelines and isolated GenAI platforms using LLMs (OLLLAMA4.0 on private clouds), MCP+RAG architectures, and agentic n8n workflows, enabling robust deployment of Python (Django, Flask) and Next.js server-side applications across AWS, Azure, and GCP.
- Implemented cloud-native architectures with GKE for containerized microservices and REST APIs, ensuring efficient model integration with enterprise platforms and streamlining data ingestion within a BigQuery-powered lakehouse.
- Designed and developed scalable backend APIs and seamless integrations using Python frameworks (Django, Flask), optimized data pipelines for AI model inference, enabling real-time predictions on large datasets.

- **Technical Leadership & Innovation**
- Led end-to-end GenAI solution development for multi-cloud environments, from secure audio ingestion and automated transcription to LLM orchestration and intelligent summarization.
- Developing Agent App streamlining Aged Care Sales, using Google's newly announced Agent-to-Agent (A2A) protocol for inter-agent communication & Model Context Protocol (MCP) for seamless integration with external tools (web search
- Enhanced system performance and scalability by integrating API management solutions and real-time Server-Sent Events (SSE) within the MCP architecture, ensuring efficient and responsive data delivery for client applications.
- Leveraged a self-recursive Artificial Superintelligence Architecture (ASI-ARCH) on the Harrison Platform (DGX A100 systems) to autonomously innovate and optimize AI models for financial management. Enhanced algorithmic trading strategies and risk mitigation by discovering emergent market principles beyond human intuition.
- Integrated M365 environments (O365, MS Teams, SharePoint online, Dynamics 365, Power Automate, Power App, Power BI) & AI Builder for D365 into the Azure Landing Zones for a comprehensive and secure platform for Microsoft workloads

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov Central Bank: Senior AI/ML Engineer/Automation Engineer/Snr DevOps/Data Engineer /AI/ML/DL/Mulesoft Developer
Responsibilities:

- **AI/ML-Powered Automation and Insights**
- Engineered end-to-end intelligent automation solutions by developing scalable Python applications that leverage NLP, computer vision, and speech-to-text models.
- Led the full lifecycle of these solutions, from audio ingestion and transcription to intelligent summarization and sentiment analysis, ensuring seamless data flow across multi-cloud environments.
- E.g: model.predict(vectorizer.transform(["RBA interest rate hike triggers AUD volatility"]))
- Developed self-learning Multi-agent systems (MAS), a subfield of distributed AI in Python, to automate and optimize algorithmic strategies, enabling models to autonomously adapt to dynamic financial market patterns.
- Developed computer vision models using OpenCV & Scikit-image to integrate image recognition for automated validation of property documents & anomaly detection, streamlining regulatory workflows & enhancing decision-making accuracy.
- Engineered intelligent rule-based Robotic Process Automation (RPA) workflows powered by NLP libraries (e.g., NLTK, spaCy) for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Integrated speech-to-text APIs like Google Cloud Speech-to-Text for automatic high-volume call transcription and sentiment analysis, directly logging customer sentiment into CRM workflows.
- Built robust predictive models and secure machine learning APIs using diverse algorithms (KNN, CNN, RNN, ANN, GAN) and ML paradigms (supervised, unsupervised, reinforcement learning), applying Scikit-learn and Keras to drive next best action (NBA) recommendations and pattern discovery.
- Leveraged PyTorch and TensorFlow on Nvidia DGX-1 systems to develop and deploy deep learning models, including fine-tuning pre-trained models for NLP and analytics, to enhance algorithmic trading strategies and risk mitigation.
- Enhanced algorithmic trading strategies and risk mitigation by discovering new market principles beyond human intuition.
- Integrated databases like PostgreSQL or MongoDB for real-time and context-aware enterprise insights.
- DevOps & Cloud Infrastructure
- Engineered scalable, mission-critical production applications and robust backend APIs in Python and MuleSoft Anypoint Platform, driving system integrations and continuous deployment for machine learning services.
    1. E.g. 1: requests.post(f"{api_url}/api/v1/deployments", json={'app':'ml-service', 'version':'v1.0.0'})
    2. E.g. 2: POST /api/v1/orders HTTP/1.1 { "item": "widget", "qty": 2 }
- Specialized in designing and implementing end-to-end automation solutions using Python & Red Hat Ansible Automation Platform (including Ansible Tower and AWX), developing modules and plugins to streamline secure, low-latency infrastructure provisioning for RBA trading platforms across hybrid cloud, on-prem, and RHEL9 environments.
  E.g.: ansible-playbook deploy_low_latency_trading_engine.yml
- Automated the creation of secure, scalable IaC CI/CD pipelines and isolated machine learning platforms, enabling robust deployment of applications across GCP, AWS, and Azure. Used Google Cloud services (GKE, Compute Engine, Cloud Functions) to ensure high availability and streamlined deployment workflows.
- Used Google Cloud Platform (GCP) infrastructure (Compute Engine, Cloud Storage, Cloud SQL, Cloud Functions, Vertex AI) to design, build, and maintain scalable, secure systems for hosting advanced machine learning models.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr ML Engineer/Automation Engineer /Integration Solution Architect /Python/ML/DL/Mulesoft Developer
Responsibilities:

- **Automated Systems and Machine Learning**
- Engineered end-to-end intelligent automation solutions by developing scalable Python applications that leverage NLP, computer vision, and speech-to-text models.

- Led the full lifecycle of these projects, from data collection and automated transcription to statistical analysis and intelligent summarization.
- Engineered self-learning Multi-agent systems (MAS) in Python to automate the management of student enrollment and financial data, enabling models to autonomously adapt to complex patterns for enhanced system performance.
- Developed computer vision models (using OpenCV and Scikit-image) to integrate image recognition for the automated analysis of visual data and anomaly detection, streamlining data processing and enhancing decision-making accuracy.
- Engineered intelligent rule-based Robotic Process Automation (RPA) workflows powered by NLP libraries (e.g., NLTK, spaCy) for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Built robust predictive models and secure APIs using diverse ML algorithms (KNN, CNN, RNN, ANN, regression, decision trees) and paradigms.
- Applied Scikit-learn and Keras for projects focused on pattern discovery and predictive analytics.Leveraged cutting-edge deep learning frameworks like Theano and Caffe to develop deep learning models, including fine-tuning pre-trained models for NLP and image recognition tasks.
- Utilized GPU-accelerated systems to develop and optimize machine learning models for research projects. Developed and tested algorithmic strategies by applying emerging principles in statistical analysis.
- Developed an intelligent web scraper (Python, requests, BeautifulSoup, pandas) with a simple UI built using Django or Flask to demonstrate robust data acquisition and processing capabilities.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- **Systems Engineering & Cloud Infrastructure**
- Designed and developed scalable backend APIs and integrations using Python frameworks (Django, Flask), optimizing data pipelines for efficient machine learning model training and enabling near real-time predictions on datasets.
- Automated the creation of secure and scalable infrastructure and CI/CD pipelines (Jenkins) for machine learning experimentation and continuous application delivery. Deployed Python applications across cloud platforms like GCP, AWS, and Azure.
- Implemented cloud-based architectures with Google Kubernetes Engine (GKE) for containerized microservices and REST APIs, ensuring efficient model integration and streamlined data ingestion.
- Used Google Cloud Platform (GCP) infrastructure (Compute Engine, Cloud Storage, and Cloud SQL) to design, build, and maintain scalable systems for hosting ML models.
- Integrated databases like PostgreSQL or MongoDB for managing and analyzing data in projects, enabling real-time and context-aware insights.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: ML Engineer /Automation Engineer/Software Test Engineer /DevOps/Data Engineer
Responsibilities:

- **Application Development and Machine Learning**
- Developed self-learning multi-agent systems in Python to automate and optimize the processing of injury claims, enabling autonomous adaptation to claims data and improving overall system efficiency.
- Python Application Developer with a robust quantitative foundation, adept at building high-performance solutions. Proficient in Python and Java, with capabilities to design, develop, and optimize ML applications and deliver actionable insights.
- Developed and deployed predictive models using Python libraries like scikit-learn to accurately assess the likelihood of specific claim outcomes or potential fraud indicators, enabling proactive risk management and optimized resource allocation in the injury compensation domain.
- Back-End Development: Engineered secure and scalable backend services using Python (e.g., Flask) and Java to efficiently manage high volumes of injury compensation claim data, ensuring data integrity and seamless integration with critical EML systems.
- Front-End Development: Designed and implemented user interfaces with jQuery and JavaScript, optimizing the claimant experience for EML's injury compensation portal and streamlining data presentation for complex claim workflows.
- Extensive expertise in programming languages such as Java, Python, C#, and PowerShell scripting languages.
- **Cloud & Data Engineering**
- Architected and implemented data solutions on AWS, focusing on traditional data warehousing and ETL processes.
- **Batch laye**r: Utilized Amazon EMR, a managed service for running Apache Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed AWS Redshift for near-real-time data analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Designed and developed data solutions using AWS services like EC2, RDS, and S3.

- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created mapping documents that defined how source data fields corresponded to target data fields.
- Implemented mapping logic based on the mapping document, including writing code to map the source data to the target data according to defined rules.
- **DevOps & Automation**
- Automated deployment of scalable machine learning applications and services, leveraging AWS services like EC2 to ensure high availability.
- Championed continuous integration and delivery (CI/CD) pipelines using tools like Jenkins for secure, continuous application delivery.
- Utilized PowerShell scripting to automate system configurations, testing, and deployment processes.
- Followed software development best practices for versioning, deployment, testing, and automation.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps/Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Software Development and Deployment: Architected and built scalable web applications using Python and deployed them on traditional servers.
- Infrastructure Management: Managed infrastructure using scripting languages and automation tools.
- Application Development: Developed applications using Python for data processing and business logic.
- Data Management: Utilized relational databases for data storage and retrieval.
- System Integration: Implemented APIs and integrated with diverse systems.
- Automated build and deployment processes using Jenkins and shell scripting, managing server configurations with Puppet to ensure reliable delivery of machine learning models for research projects.
- Deployed and maintained web applications on virtual servers. Managed infrastructure using shell scripts and other automation tools, integrating them into CI/CD pipelines (e.g., Jenkins). Developed REST APIs using Python and Java.
- Monitored server resources using on-premise tools and implemented logging, monitoring, and security measures.
- **Security and Compliance**: Implemented security best practices and regularly patched systems and applications.
- **Agile and Data Expertise**: Experienced working in agile environments with strong data engineering practices.
- Other Proficiencies: Possessed a strong understanding of SQL, Python scripting, and SharePoint.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of continuous integration (CI) and automated deployment processes, leading to a significant reduction in software feature delivery time.
- Utilized scripting languages and configuration management tools to implement Infrastructure as Code (IaC) principles, enhancing deployment processes and scalability.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Administered and monitored Unix/Linux-based systems, ensuring the smooth operation of critical applications.
- Provided Level 2/3 technical support for all Microsoft infrastructures, including Active Directory (AD), DNS, DHCP, Exchange 2010, and SharePoint.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running costs.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Served as Project Manager to replace the existing insurance web application (from IBM AS/400 and LAMP) with a WISA-based system, significantly reducing IT operational costs and integrating with a Salesforce Automation (SFA) platform for improved marketing results.
- Planned, implemented, and supported all core IT systems including Hyper-V, VMware ESX & vSphere 4.x, HP/EMC/Dell SAN Storage, and Windows Server 2008, and applied security principles and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for desktop productivity suite issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for Business Intelligence (BI) solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized C++, C#, VB.NET, and T-SQL for SSIS and SSRS programming.
- Designed data warehouse architecture, including hardware infrastructure, schema definitions, and Fact and Relational Dimensions tables.
- Developed data warehouse and business reporting solutions to meet stakeholder requirements using SSIS, SSAS, and SSRS.
- Used SSIS ETL tools to extract and transform complex data, loading it into the data warehouse to produce Analysis Services Cubes for business reporting via SSRS or Data Marts to serve business applications.
- Created and deployed reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio, including debugging, monitoring, and troubleshooting BI solutions.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Managed and maintained company databases, ensuring data accuracy and integrity.
- Defined the target warehouse module, and created target user, module, and schema.
- Built and deployed machine learning models using k-nearest neighbors (k-NN) for customer churn prediction and fraud detection, achieving a 15% improvement in accuracy.
- Involved in UAT and TEST in a QA development environment by writing VBScript and batch files to automate processes.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI (ChatGPT-4o, Multi-GPTs, Auto-Agents), LM-Studio, Azure OpenAI ML Services, Copilot, Google GEMINI, MakerSuite
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
- **Workflow & Automation Platforms**: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve

AI Engineer – Altitude ->InfoSys Consulting → Westpac – 22/8/2025 – Interviewed on 25/8/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior AI Engineer-Python / GenAI | Greenfield AI Build | Enterprise Innovation /Automation Engineer/Full Stack Develop/r Software Engineer/Snr DevOps /Data Engineer /Mulesoft /AI/ML/DL/BI/SP/D365 Developer -Japan (7.5yrs) and Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior AI Engineer- Python / GenAI | Greenfield AI Build | Enterprise Innovation /Automation Engineer
Responsibilities:

- **Agentic Platform Development & Generative AI**
- Architected and deployed enterprise-grade Generative AI solutions utilizing advanced agentic systems for research, writing, editing, and publishing. The systems, powered by LLMs like GPT, Claude, and Gemini and enhanced with Retrieval-Augmented Generation (RAG), were built to automate business processes and drive stakeholder engagement.
- Developed sophisticated agentic AI systems for autonomous task execution, with a focus on integrating a Semantic Kernel for orchestrating complex functions and enabling advanced semantic research to inform real-time, data-driven decisions.
- Developed advanced agentic applications leveraging cutting-edge agentic frameworks including LangGraph, AutoGen, CrewAI, and Semantic Kernel for sophisticated LLM orchestration and autonomous AI agent development.
- Architected and implemented AI agent architecture using LangChain/TensorFlow to build self-learning systems that adapt based on customer transaction patterns, significantly increasing automation and system performance.
- Utilized inter-agent communication protocols and seamless integration with external tools (e.g., web search APIs) within these agentic applications, further operationalizing through n8n for workflow automation to drive enhanced operational efficiency, automation, and intelligent decision-making.
- Built and optimized Retrieval-Augmented Generation (RAG) pipelines for enhanced contextual understanding and response generation, focusing on pre-retrieval optimization through intelligent chunking and metadata enrichment to improve retrieval accuracy. E.g., text_splitter = RecursiveCharacterTextSplitter(chunk_size=1000, chunk_overlap=200)
- Integrated advanced Graph RAG techniques to improve semantic search and knowledge retrieval, creating knowledge graphs from unstructured data to enable more complex and precise queries that go beyond vector similarity.
  E.g., kg_builder = SimpleKGPipeline(llm=ex_llm, driver=neo4j_driver, ...)
- Developed an AI-powered insights platform for a major financial institution by creating a knowledge graph to map complex customer and transaction data. Used embeddings to represent this information as high-dimensional vectors, which were then stored in a vector database (Pinecone/Weaviate) to enable lightning-fast semantic search and relationship discovery, enhancing fraud detection and personalized financial recommendations.
- Integrated MongoDB Atlas as a vector database for RAG (Retrieval-Augmented Generation), enabling real-time and context-aware enterprise insights.
- **AI-Powered Automation & Insights**
- **AI-Augmented RPA with Natural Language Processing (NLP)**: Engineered intelligent RPA workflows powered by NLP for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Developed scalable Python applications, orchestrating LangChain (for LLM orchestration), LlamaIndex (for data indexing), and Haystack (for production-ready search and QA). Integrated Amazon Transcribe Call Analytics for automatic high-volume call transcription and sentiment analysis, directly logging customer sentiment into CRM workflows.
- Developed scalable Python applications, leveraging Generative AI frameworks (LangChain for LLM orchestration, LlamaIndex for RAG) and speech-to-text models like Whisper to automate the transcription, summarization, and analysis of customer calls. Optimized deployment of these systems on AWS (using Bedrock and OpenAI frameworks) while integrating with APIs, microservices, and enterprise data platforms.
- Developed an intelligent AI agent using an Amazon Scraper (Python, requests, BeautifulSoup, pandas) with Streamlit UI, demonstrating robust data acquisition and constraints bypassing.
- **Machine Learning & Model Development**
- Leveraged PyTorch, TensorFlow, and Hugging Face Transformers to develop and deploy LLMs (BERT, GPT-4o, Jurassic-1, Bloom) and deep learning models, including fine-tuning pre-trained models for NLP and analytics.
- Built robust predictive models and secure AI APIs using diverse machine learning algorithms (KNN, CNN, RNN, ANN, GAN, Recurrent Neural Network (RNN) units (like LSTM and GRU)
- regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning), applying Scikit-learn and Keras to drive Next Best Action (NBA) recommendations and pattern discovery.
- Drove the full lifecycle of AI solutions, from ideation and opportunity identification with key stakeholders to the design and delivery of scalable prototypes. Conducted comprehensive experiments to evaluate model performance, iterating on design to optimize outcomes and ensure compliance with ethical AI standards.
- **DevOps, Cloud Infrastructure & Backend Development**
- Pioneered the use of Large Language Models (LLMs) as optimizers within CI/CD pipelines to dynamically generate, test, and refactor code, leading to a 25% reduction in deployment time and a significant improvement in code quality across the team. E.g., llm_output = gemini.generate_text(f"Refactor this code to optimize performance: {code_snippet}")
- Designed and implemented end-to-end automation solutions using Python and Ansible (Tower, AWX, Red Hat AAP 2.5), developing custom modules and plugins to streamline infrastructure provisioning and configuration across hybrid cloud, on-prem, and Red Hat Enterprise Linux (RHEL9/10) environments. E.g., ansible-playbook provision_rhel_app_servers.yml
- Automated the deployment of scalable GenAI applications and microservices across hybrid cloud environments, leveraging cloud-native AWS services like ECS/EKS, Lambda, & SageMaker to ensure high availability & streamlined CI/CD pipelines.
- Automated creation of secure, scalable IaC CI/CD pipelines and isolated GenAI platforms using LLMs (OLLLAMA4.0 on private clouds), MCP+RAG architectures, and agentic n8n workflows, enabling robust deployment of Python (Django, Flask) and Next.js server-side applications across AWS, Azure, and GCP.

- Leveraged AWS Kiro's spec-driven development & agentic capabilities to design, build, and maintain scalable, secure AWS infrastructure (EC2, EKS, S3, RDS, Lambda, SageMaker, Amazon Neptune) for hosting advanced Graph RAG pipelines, vector databases, & LLMs, Using AWS Bedrock for diverse foundational models (e.g., Amazon, Anthropic, AI21 Labs).
- Implemented cloud-native architectures with AWS ECS/EKS for containerized microservices and REST APIs, ensuring efficient model integration with enterprise platforms streamlined data ingestion within an AWS Redshift-powered lakehouse.
- Designed and developed scalable backend APIs and seamless integrations using Python frameworks (Django, Flask), optimized data pipelines for AI model inference, enabling real-time predictions on large datasets.
- Engineered scalable, mission-critical production applications in Python, driving robust system integrations and continuous deployment for AI services with core code such as `requests.post(f"{api_url}/api/v1/deployments", json={'app':'genai-service', 'version':'v1.0.0'})`.
- **Technical Leadership & Innovation**
- Led end-to-end GenAI solution development from secure audio ingestion, automated transcription, LLM orchestration, RAG pipeline integration, to intelligent summarization—ensuring seamless data flow across multi-cloud.
- Leveraged a self-recursive Artificial Superintelligence Architecture (ASI-ARCH) on the Harrison Platform (Nvidia DGX-1 systems) to autonomously innovate and optimize AI models for financial management. Enhanced algorithmic trading strategies and risk mitigation by discovering emergent market principles beyond human intuition.
- Integrated M365 environments (O365, MS Teams, SharePoint online, Dynamics 365, Power Automate,Power App, Power BI) & AI Builder for D365 into the AWS Landing Zones for a comprehensive and secure platform for Microsoft workloads.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior AI/ML Engineer/Automation Engineer/Snr DevOps/Data Engineer /AI/ML/DL/Mulesoft Developer
Responsibilities:

- AI/ML/DL-Powered Automation & Insights
- Developed self-learning Multi-agent systems (MAS), a subfield of distributed AI in Python to automate and optimize algorithmic strategies, enabling models to autonomously adapt to dynamic financial market patterns.
- Developed computer vision models using OpenCV & Scikit-image to integrate image recognition for automated validation of property documents & anomaly detection, streamlining regulatory workflows & enhancing decision-making accuracy.
- Engineered intelligent rule-based Robotic Process Automation (RPA) workflows powered by NLP libraries (e.g., NLTK, spaCy) for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Developed scalable Python applications, leveraging frameworks like scikit-learn and NLTK for data indexing and natural language processing. E.g., model.predict(vectorizer.transform(["RBA interest rate hike triggers AUD volatility"]))
- Integrated speech-to-text APIs (like early AWS Transcribe, though this was limited at the time, or other third-party services) for automatic high-volume call transcription & sentiment analysis, directly logging customer sentiment into CRM workflows.
- **AI/ML Solution Development**: Led end-to-end machine learning solution development from secure audio ingestion, automated transcription, to intelligent summarization—ensuring seamless data flow across multi-cloud.
- **Technical Foundations & Modeling**
- Built robust predictive models and secure machine learning APIs using diverse algorithms (KNN, CNN, RNN, ANN, GAN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning), applying Scikit-learn and Keras to drive next best action (NBA) recommendations and pattern discovery.
- Deep Learning Models: Leveraged PyTorch and TensorFlow to develop and deploy deep learning models, including fine-tuning pre-trained models for NLP and analytics.
- Utilized Nvidia DGX-1 systems to develop and optimize machine learning models for financial management. Enhanced algorithmic trading strategies and risk mitigation by discovering new market principles beyond human intuition.
- Engineered scalable, mission-critical production applications in Python, driving robust system integrations and continuous deployment for machine learning services with core code such as requests.post(f"{api_url}/api/v1/deployments", json={'app':'ml-service', 'version':'v1.0.0'}).
- Developed and deployed scalable applications using MuleSoft Anypoint Platform, including Mule 4 & Anypoint Studio 7.x, with a strong focus on robust backend API development and seamless system integrations.
    1. API Development: POST /api/v1/orders HTTP/1.1 { "item": "widget", "qty": 2 }
    2. System Integration: Flow orchestrates Salesforce, SAP, and custom REST services for order processing.
- **DevOps & Cloud Infrastructure**
- Specialized in designing and implementing end-to-end automation solutions using Python & Red Hat Ansible Automation Platform (including Ansible Tower and AWX), developing modules and plugins to streamline secure, low-latency infrastructure provisioning for RBA trading platforms across hybrid cloud, on-prem, and RHEL9 environments. E.g., ansible-playbook deploy_low_latency_trading_engine.yml to automate trading engine deployments
- Automated the deployment of scalable machine learning applications and microservices across hybrid cloud environments, leveraging cloud-native AWS services like EC2, ECS, and Lambda to ensure high availability & streamlined CI/CD pipelines.
- Automated the creation of secure, scalable IaC CI/CD pipelines and isolated machine learning platforms, enabling robust deployment of Python (Django, Flask) and other server-side applications across AWS, Azure, and GCP.
- Implemented cloud-native architectures with AWS ECS for containerized microservices and REST APIs, ensuring efficient model integration with enterprise platforms and streamlined data ingestion within a data lakehouse.

- AWS Infrastructure: Used AWS infrastructure (EC2, S3, RDS, Lambda, SageMaker) to design, build, and maintain scalable, secure systems for hosting advanced machine learning models.
- Data Integration: Integrated databases like PostgreSQL or MongoDB for real-time and context-aware enterprise insights.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr ML Engineer/Automation Engineer /Integration Solution Architect /ML/DL/Mulesoft Developer
Responsibilities:

- **Automated Systems and Machine Learning**
- Engineered self-learning Multi-agent systems (MAS), a subfield of distributed AI in Python to automate the management of student enrollment and financial data, enabling models to autonomously adapt to complex payment and enrollment patterns for enhanced system performance.
- Developed computer vision models (using OpenCV and Scikit-image) to integrate image recognition for the automated analysis of visual data and anomaly detection, streamlining data processing and enhancing decision-making accuracy.
- Engineered intelligent rule-based Robotic Process Automation (RPA) workflows powered by NLP libraries (e.g., NLTK, spaCy) for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- **Machine Learning and Data Insights**
- Led end-to-end machine learning project development, from data collection and automated transcription to statistical analysis and intelligent summarization, ensuring efficient data flow for research and academic projects.
- Developed scalable Python applications, leveraging core frameworks like scikit-learn and NLTK for data preprocessing, modeling, and natural language processing. Utilized third-party speech-to-text services for automatic transcription and sentiment analysis, integrating results into custom-built data analysis pipelines.
- Leveraged cutting-edge deep learning frameworks like Theano and Caffe to develop deep learning models, including fine-tuning pre-trained models for NLP and image recognition tasks.
- Built robust predictive models and secure APIs using diverse machine learning algorithms (KNN, CNN, RNN, ANN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning). Applied Scikit-learn and Keras for projects focused on pattern discovery and predictive analytics.
- Used AWS infrastructure (EC2, S3, and RDS) to design, build, and maintain scalable systems for hosting ML models.
- Integrated databases like PostgreSQL or MongoDB for managing and analyzing data in projects, enabling real-time and context-aware insights.
- Implemented cloud-based architectures with AWS ECS for containerized microservices and REST APIs, ensuring efficient model integration and streamlined data ingestion.
- Developed an intelligent web scraper (Python, requests, BeautifulSoup, pandas) with a simple UI built using Django or Flask to demonstrate robust data acquisition and processing capabilities.
- **Systems Engineering and Automation**
- Utilized GPU-accelerated systems to develop and optimize machine learning models for research projects. Developed and tested algorithmic strategies by applying emerging principles in statistical analysis.
- Designed and developed scalable backend APIs and integrations using Python frameworks (Django, Flask), optimizing data pipelines for efficient machine learning model training and enabling near real-time predictions on datasets.
- Automated the deployment of machine learning applications and services, experimenting with early cloud-native AWS services like EC2, ECS, and Lambda to ensure high availability and streamlined deployment workflows.
- Applied Infrastructure-as-Code (Terraform) and automated CI/CD pipelines (e.g., Jenkins) for secure and continuous application delivery across various environments.
- Automated the creation of secure and scalable infrastructure for machine learning experimentation, enabling robust deployment of Python (Django, Flask) and other server-side applications across cloud platforms like AWS, Azure, and GCP.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: ML Engineer /Automation Engineer/Software Test Engineer /DevOps/Data Engineer
Responsibilities:

- **Application Development and Machine Learning**
- Developed self-learning multi-agent systems in Python to automate and optimize the processing of injury claims, enabling autonomous adaptation to claims data and improving overall system efficiency.
- Python Application Developer with a robust quantitative foundation, adept at building high-performance solutions. Proficient in Python and Java, with capabilities to design, develop, and optimize ML applications and deliver actionable insights.
- Developed and deployed predictive models using Python libraries like scikit-learn to accurately assess the likelihood of specific claim outcomes or potential fraud indicators, enabling proactive risk management and optimized resource allocation in the injury compensation domain.

- Back-End Development: Engineered secure and scalable backend services using Python (e.g., Flask) and Java to efficiently manage high volumes of injury compensation claim data, ensuring data integrity and seamless integration with critical EML systems.
- Front-End Development: Designed and implemented user interfaces with jQuery and JavaScript, optimizing the claimant experience for EML's injury compensation portal and streamlining data presentation for complex claim workflows.
- Extensive expertise in programming languages such as Java, Python, C#, and PowerShell scripting languages.
- **Data Engineering & Architecture**
- Cloud & Data Engineering: Architected and implemented data solutions on AWS, focusing on traditional data warehousing and ETL processes.
- Batch layer: Utilized Amazon EMR, a managed service for running Apache Hadoop clusters, for batch processing of large data volumes.
- Speed layer: Employed AWS Redshift for near-real-time data analysis.
- Designed and developed data solutions using AWS services like EC2, RDS, and S3.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created mapping documents that defined how source data fields corresponded to target data fields.
- Implemented mapping logic based on the mapping document, including writing code to map the source data to the target data according to defined rules.
- **DevOps & Automation**
- Automated deployment of scalable machine learning applications and services, leveraging AWS services like EC2 to ensure high availability.
- Championed continuous integration and delivery (CI/CD) pipelines using tools like Jenkins for secure, continuous application delivery.
- Utilized PowerShell scripting to automate system configurations, testing, and deployment processes.
- Followed software development best practices for versioning, deployment, testing, and automation.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps/Data Engineer /Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Software Development and Deployment: Architected and built scalable web applications using Python and deployed them on traditional servers.
- Infrastructure Management: Managed infrastructure using scripting languages and automation tools.
- Application Development: Developed applications using Python for data processing and business logic.
- Data Management: Utilized relational databases for data storage and retrieval.
- System Integration: Implemented APIs and integrated with diverse systems.
- Automated build and deployment processes using Jenkins and shell scripting, managing server configurations with Puppet to ensure reliable delivery of machine learning models for research projects.
- Deployed and maintained web applications on virtual servers. Managed infrastructure using shell scripts and other automation tools, integrating them into CI/CD pipelines (e.g., Jenkins). Developed REST APIs using Python and Java.
- Monitored server resources using on-premise tools and implemented logging, monitoring, and security measures.
- **Security and Compliance**: Implemented security best practices and regularly patched systems and applications.
- **Agile and Data Expertise**: Experienced working in agile environments with strong data engineering practices.
- Other Proficiencies: Possessed a strong understanding of SQL, Python scripting, and SharePoint.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of continuous integration (CI) and automated deployment processes, leading to a significant reduction in software feature delivery time.
- Utilized scripting languages and configuration management tools to implement Infrastructure as Code (IaC) principles, enhancing deployment processes and scalability.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Administered and monitored Unix/Linux-based systems, ensuring the smooth operation of critical applications.
- Provided Level 2/3 technical support for all Microsoft infrastructures, including Active Directory (AD), DNS, DHCP, Exchange 2010, and SharePoint.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running costs.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH

Responsibilities:

- Served as Project Manager to replace the existing insurance web application (from IBM AS/400 and LAMP) with a WISA-based system, significantly reducing IT operational costs and integrating with a Salesforce Automation (SFA) platform for improved marketing results.
- Planned, implemented, and supported all core IT systems including Hyper-V, VMware ESX & vSphere 4.x, HP/EMC/Dell SAN Storage, and Windows Server 2008, and applied security principles and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for desktop productivity suite issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH**
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for Business Intelligence (BI) solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized C++, C#, VB.NET, and T-SQL for SSIS and SSRS programming.
- Designed data warehouse architecture, including hardware infrastructure, schema definitions, and Fact and Relational Dimensions tables.
- Developed data warehouse and business reporting solutions to meet stakeholder requirements using SSIS, SSAS, and SSRS.
- Used SSIS ETL tools to extract and transform complex data, loading it into the data warehouse to produce Analysis Services Cubes for business reporting via SSRS or Data Marts to serve business applications.
- Created and deployed reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio, including debugging, monitoring, and troubleshooting BI solutions.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Managed and maintained company databases, ensuring data accuracy and integrity.
- Defined the target warehouse module, and created target user, module, and schema.
- Built and deployed machine learning models using k-nearest neighbors (k-NN) for customer churn prediction and fraud detection, achieving a 15% improvement in accuracy.
- Involved in UAT and TEST in a QA development environment by writing VBScript and batch files to automate processes.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI (ChatGPT-4o, Multi-GPTs, Auto-Agents), LM-Studio, Azure OpenAI ML Services, Copilot, Google GEMINI, MakerSuite
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
- **Workflow & Automation Platforms**: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve